

Hematologic Aspects of Kidney Disease

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CHAPTER OUTLINE

ANEMIA OF KIDNEY DISEASE, 1565
DISORDERS OF HEMOSTASIS IN CHRONIC KIDNEY DISEASE, 1590

WHITE CELL FUNCTION IN CHRONIC KIDNEY DISEASE, 1595

KEY POINTS

- Anemia is a hallmark of chronic kidney disease (CKD), which profoundly affects a patient's well-being and is associated with adverse health outcomes.
- The prevalence of anemia begins to increase significantly with an estimated glomerular filtration rate below 60 mL/min/1.73 m², but anemia is generally not a severe complication of CKD until the glomerular filtration rate is below 30 mL/min/1.73 m².
- Anemia is a more significant problem for younger women, older patients, Black patients, and patients with diabetes mellitus.
- Serum erythropoietin (EPO) concentrations are generally equal to or lower than those in patients without CKD.
- Even with advanced CKD, the ability to produce EPO is preserved, but responsiveness to lower Hgb is attenuated.
- Major pathogenic factors of renal anemia are inadequate EPO production and absolute and functional iron deficiency.
- The two mainstays of anemia treatment are therapy with recombinant human EPO or one of its hyperglycosylated derivatives and oral or intravenous iron.
- Hypoxia-inducible prolyl-hydroxylase inhibitors stimulate endogenous EPO production and provide an orally active alternative to ESAs; however, safety concerns restrict their use.
- When treating CKD-related anemia, subnormal rather than normal hemoglobin concentrations should be targeted.

ANEMIA OF KIDNEY DISEASE

DEFINITION AND PREVALENCE OF ANEMIA IN CHRONIC KIDNEY DISEASE

Anemia is characterized as a reduced mass of red blood cells (RBCs), resulting in reduced oxygen-carrying capacity and oxygen delivery to the body's tissues.¹⁻³ Because direct measurements of red cell mass are cumbersome and not readily available, anemia is defined as a reduction below the normal range for hemoglobin (Hgb) concentration

in the blood and/or hematocrit (Hct; the percentage of whole blood volume occupied by RBCs). The World Health Organization (WHO) defines anemia in adults as a Hgb concentration below 13.0 g/dL for men and below 12.0 g/dL for women.⁴ This definition has been adopted in the clinical practice guideline for anemia in CKD developed by Kidney Disease: Improving Global Outcomes (KDIGO).^{5,6}

In individuals with CKD, the prevalence of anemia depends on a variety of factors including specifically the severity of CKD (Fig. 53.1) but also the cause of CKD, demographic characteristics, habitat, and treatment with iron and/or ESAs. Overall, a remarkable variability in the

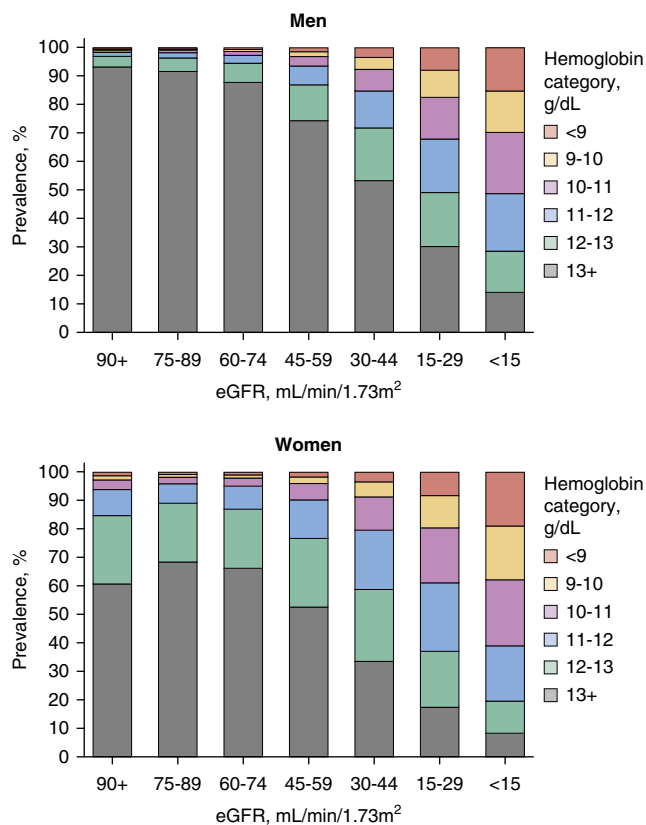


Fig. 53.1 Prevalence of anemia by estimated glomerular filtration rate (eGFR) category, stratified by sex. Outpatient data from 5,004,957 individuals across 57 health care centers in the United States from 2016 to 2019 were extracted from the Optum Labs Data Warehouse. (From Farrington DK, Sang Y, Grams ME, et al. Anemia prevalence, type, and associated risks in a cohort of 5.0 million insured patients in the United States by level of kidney function. *Am J Kidney Dis*. 2023;81(2):201–209.)

prevalence of anemia has been reported across different stages of CKD and across different countries and regions.^{6,7} Anemia has been reported in 20% to 60% of patients with CKD not receiving dialysis (ND-CKD) and in 50% to 80% of those with CKD stage G4.⁶⁻¹⁰

The interaction of aging and loss of kidney function might be expected to raise the prevalence of anemia. In fact, the relation is more complex.¹¹ Men with CKD tend to have a higher prevalence of anemia with older age, but among women with CKD, anemia is more frequent overall¹² and at a younger age.¹³ The high prevalence of iron deficiency in menstruating women likely accounts for this difference. Other studies found that age was an important determinant of anemia in the absence of CKD, whereas this effect was lost in the presence of CKD, which became the strongest determinant of anemia.^{14,15}

Anemia develops earlier in the course of CKD and tends to be more severe in patients with diabetes mellitus than patients without diabetes.¹⁶⁻²² The difference among patients with and without diabetes is most prominent in patients with CKD stage G3, in whom the prevalence of anemia was found to be nearly threefold greater in those with diabetes. Subclinical kidney disease in diabetes may contribute to compromised EPO production and anemia.

Several reports including an analysis of the United States (U.S.) National Health and Nutrition Examination Survey (NHANES) data suggest that among other factors (age ≥ 75 years, female sex, CKD stage $\geq G3b$ and concurrent diabetes), anemia is also more likely to occur in Black patients compared with patients of other races.¹⁰ Despite several factors identified to be associated with the prevalence and severity of anemia, a comparison of Hgb concentrations in more than 58,000 individuals with ND-CKD enrolled into 21 cohort studies revealed marked regional variation unexplained by demographics, estimated glomerular filtration rate (eGFR), or comorbidities.²³

Among patients on maintenance dialysis, anemia is nearly universal. According to a U.S. Renal Data System (USRDS) annual report, more than 85% and 21% of patients receiving dialysis in the United States have Hgb concentrations below 12 g/dL and 10 g/dL, respectively.²⁴ In kidney transplant recipients, the prevalence of anemia was reported to range between 20% and 50%, varying with time after transplantation.²⁵

Persons living at higher altitudes are characterized by a larger red cell mass and lower Hgb oxygen affinity, compensatory changes required to maintain tissue oxygen delivery, given the reduced ambient oxygen tension at high altitudes.²⁶⁻²⁹ This response is at least partially preserved in people with kidney disease.³⁰⁻³²

In general, Hgb concentrations in patients with CKD can be confounded by treatment with medications including ESAs and iron (see later). Liberal ESA use before the initiation of dialysis and in patients on dialysis had become more conservative after publication of the results of randomized controlled trials (RCTs) that led to safety concerns about higher Hgb targets and subsequent changes in prescribing instructions and guidelines (see later).³³ In most patients beginning dialysis in the United States, the Hgb value at dialysis initiation has since then dropped below 10 g/dL.^{34,35}

NORMAL ERYTHROPOIESIS

The delivery of oxygen to peripheral tissues is a highly regulated process; a crucially important determinant is the red blood cell mass, which is determined by the dynamic balance between the removal of older cells from the circulation and production of newer cells by the bone marrow. Under normal conditions, the most senescent (approximately 1%) circulating erythrocytes are replaced daily, corresponding to about 250 billion erythrocytes, with 2.5 to 3.0 million new erythrocytes (reticulocytes) being produced each second.³⁶ The control of red blood cell mass is based on a classic negative feedback loop and is mediated by changes in the production of the hormone EPO, a 30.4-kDa glycoprotein that is the major regulator of erythrocyte production. EPO is almost exclusively produced in kidneys and liver and regulates the production of reticulocytes by interaction with specific EPO receptors (EPO-Rs) on bone marrow erythroid progenitors. For this mechanism to function properly, several other nutrient cofactors such as iron, vitamin B₁₂, and folate are also required.

ERYTHROPOIETIN

EPO production in the kidney is modulated by the delivery of oxygen from the circulating erythrocytes. When the

circulating erythrocyte mass decreases because of decreased production, enhanced destruction, or loss of erythrocytes, the subsequent reduction in oxygen delivery to the kidneys results in increased production of this hormone. The first recognition of the linkage between hypoxia and erythrocyte quantity arose from astute 19th century observations on the effects of living at a higher altitude.^{37,38} Carnot and Deflandre³⁹ first postulated that a humoral factor (termed hemopoietin) might regulate erythropoiesis. They injected serum from anemic rabbits into normal animals and found increased reticulocyte counts. Their observation was probably an artifact because the amount of serum transferred was low, and attempts to confirm their results were unsuccessful.⁴⁰

Forty-four years later, Reissmann⁴¹ rekindled interest in the field with ingenious experiments in parabiotic rats. In this model, rats were joined by skin and muscle, ear to tail, living for 3 months in parabiosis that permitted exchange of plasma but not RBCs. When one animal breathed air with low oxygen tension and the other breathed normal air, both animals demonstrated increased bone marrow erythropoiesis, providing strong evidence that a humoral factor was the stimulus for erythropoiesis. In 1953, Erslev⁴² definitively demonstrated the erythropoietic role of the serum factor, then termed “erythropoietin.” He infused 100 to 200 mL of plasma from bled, anemic rabbits into normal rabbit recipients, and the reticulocyte count increased fourfold within 4 days.^{42,43} In 1957, Jacobson and coworkers⁴⁴ provided indirect evidence that the kidneys were the primary source of EPO. Resection of various organs did not affect EPO production after phlebotomy, but bilaterally nephrectomized rats and rabbits failed to increase EPO production (assayed by iron-59 incorporation into erythrocytes) after blood loss.⁴⁴ Further studies by Koury and Lacombe and associates^{45,46} demonstrated that the cells responsible for EPO production are peritubular interstitial cells, which were subsequently identified as peritubular fibroblasts, located in the renal cortex.⁴⁷⁻⁴⁹ Although the nature of these cells has still not been fully clarified, they have been suggested to be derived from the neural crest⁵⁰ and share characteristics of pericytes.^{51,52} More recently, it has also been postulated that these cells form a distinct cell entity.⁵³

With the increasing severity of anemia, the number of EPO-producing peritubular cells increases. This recruitment is an all-or-none induction of individual EPO-producing cells mostly in the inner cortex but extending throughout the renal cortex when the anemia is severe.^{54,55} EPO production is regulated by a specific hypoxia-sensing mechanism based on transcription factors stabilized by hypoxia, called “hypoxia-inducible factors” (HIFs; Fig. 53.2).⁵² This regulatory mechanism is not unique to EPO and is based on the capability of two separate helix loop–helix components, HIF- α and HIF- β , to bind as a complex to specific hypoxia-responsive DNA elements, which regulate the transcription of hypoxia-inducible genes.⁵⁶ The concentrations of the β subunit do not respond to hypoxia.^{57,58} The α subunits (1 α , 2 α , and 3 α) are produced constitutively, but they are rapidly degraded in the presence of oxygen by the ubiquitin-proteasome system.⁵⁹⁻⁶² In hypoxic conditions, degradation of the α subunits is inhibited, leading to rapid increases in HIF- α concentrations and to the formation of the HIF

transcription complex. For EPO regulation, HIF-2 appears to be the important HIF isoform.⁶³⁻⁶⁸ Renal HIF-2 is required for hypoxia-driven EPO production; in the absence of renal HIF-2, hepatic HIF-2 becomes the main regulator of EPO production.^{52,63} HIF-2 α , together with HIF- β , hepatocyte nuclear factor-4 (HNF-4), and p300 bind to a 120-bp enhancer, which is located at the 3' end of the human EPO polyadenylation signal.^{36,58,69-71} This interaction and HIF binding in the 5' region flanking region results in rapid EPO transcription, followed by translation and secretion of the EPO glycoprotein.⁷²⁻⁷⁵

The rapid degradation of HIF- α in the presence of oxygen depends on binding of the tumor suppressor protein von Hippel-Lindau (VHL), a process that results in tagging of the molecule for proteasomal degradation via polyubiquitination by ubiquitin ligase.^{76,77} This regulatory mechanism is based on the hydroxylation of two proline residues, which are critical for recognition of HIF- α by VHL. An additional hydroxylation of one asparagine residue is required for HIF binding with p300.⁷⁸⁻⁸² Hydroxylation at these three sites depends on the presence of oxygen as a molecular substrate for specific hydroxylase enzymes, placing these enzymes in a central role for sensing oxygen and detecting hypoxia. In 2019, the Nobel Prize for Medicine or Physiology was awarded to G. Semenza, P.J. Ratcliffe, and B. Kaelin for their discovery of the oxygen-sensing mechanism.⁸³

A mutation in the VHL protein that impairs the degradation of HIF- α and increases EPO production causes Chuvash congenital polycythemia, an autosomal recessive disorder that is endemic in the mid-Volga River region.⁸⁴ HIF-2 α mRNA but not HIF-1 α mRNA displays a typical iron response element (IRE) in its 5' untranslated region (UTR), which constitutes an important regulatory loop for HIF-2 α messenger RNA (mRNA) translation in the presence of hypoxia or iron loading.⁸⁵⁻⁸⁷ In conditions of reduced iron availability, this regulatory mechanism allows iron response protein 1 (IRP1) to bind with high affinity to the IRE, thereby inhibiting mRNA translation, and decreasing HIF-2 α synthesis and EPO production. When cellular iron is abundant, IRP1 loses its RNA binding activity and functions as the cytosolic enzyme aconitase, resulting in derepression of mRNA translation, which increases HIF-2 α synthesis and EPO production.

Using multiple isolation steps, Miyake and colleagues⁸⁸ purified a small quantity of pure EPO glycoprotein from 2550 L of urine from patients with aplastic anemia. This purified EPO allowed successful cloning of the gene in 1985 by Lin and associates.⁸⁹ They found that the gene encodes a protein of 193 amino acids, including a 27-amino acid leader sequence and a terminal single amino acid that are cleaved during processing, resulting in a 165-amino acid mature EPO molecule. When these investigators introduced the gene into Chinese hamster ovary cells, EPO with full biologic activity was produced.⁸⁹ These findings were confirmed by an almost simultaneous report by Jacobs and colleagues.⁹⁰ The results led in short order to the development of techniques to produce recombinant human EPO (rhEPO). By 1989, clinical trials of rhEPO had demonstrated its remarkable efficacy,⁹¹⁻⁹⁴ leading to regulatory approval and routine clinical use of rhEPO in patients with anemia in the setting of kidney failure.

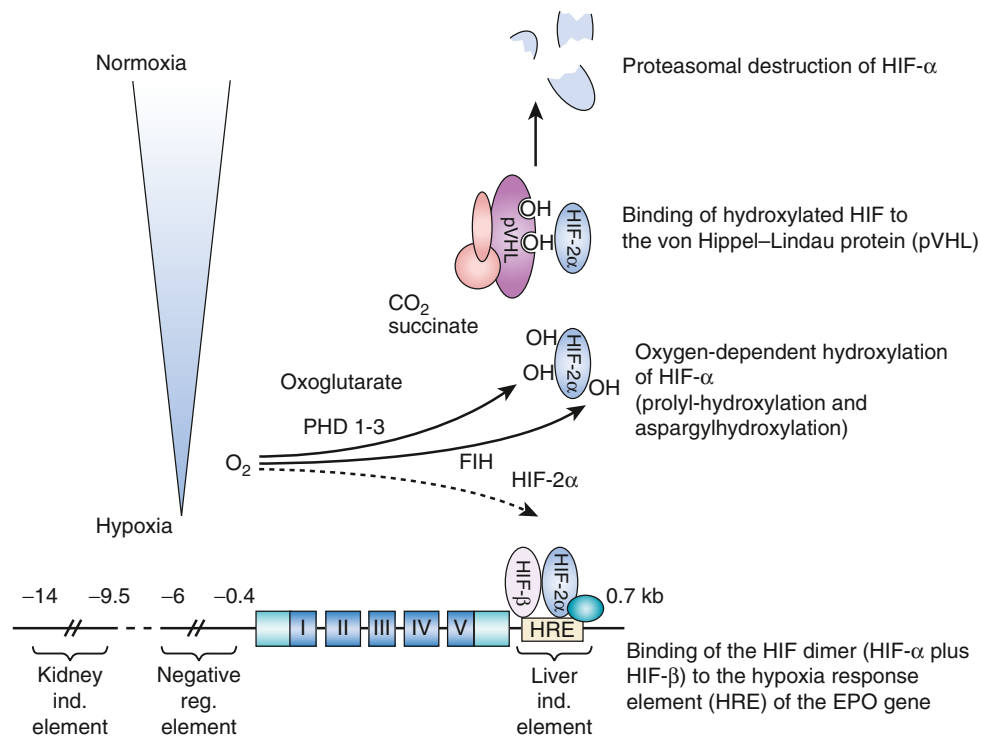


Fig. 53.2 Schematic presentation of hypoxia-inducible factor (HIF) signaling and the oxygen-dependent control of erythropoietin (EPO) gene expression. HIF consists of one of two oxygen-dependent α subunits, HIF-1 α and HIF-2 α , and a constitutive β subunit. For EPO regulation, HIF-2 α is the relevant isoform. In the presence of oxygen (normoxia), HIF- α is hydroxylated at two prolyl and one asparagyl residues through prolyl hydroxylases (PHDs 1–3) and an asparagyl hydroxylase (factor-inhibiting HIF [FIH]), enzymes that require oxoglutarate as a cosubstrate. Hydroxylation of the asparagyl residue inhibits binding of the transcriptional coactivator p300, and hydroxylation of the prolyl residues enables binding to the von Hippel-Lindau protein, which represents the recognition component of an E3 ubiquitin ligase. Thus, hydroxylated HIF- α is targeted for proteasomal destruction, and hydroxylated HIF that escapes destruction is not transcriptionally active. Under hypoxia, there is limited substrate (oxygen) for the hydroxylation reactions, and thus HIF- α is stabilized, can bind to hypoxia-responsive elements of its target genes, and can induce or enhance their transcription. The hypoxia response element (HRE) of the EPO gene in the liver is located 5' of the gene, while the HRE in the kidney is located 3'; other regulatory elements determine the tissue specificity, thereby limiting EPO expression mainly to liver and kidneys. *ind.*, Inducible; *reg.*, regulatory.

EPO is a member of the family of class 1 cytokines.⁹⁵ The carbohydrate moiety is important for molecular stability, whereas the 165-amino acid protein component is critical for receptor binding.^{96,98} The four discrete carbohydrate chains, three N-linked and one O-linked, each have two to four branches, and most terminate in a negatively charged sialic acid.^{96,99-103} The physiologic role of the carbohydrate chains is complex in that they are required for in vivo biologic activity of EPO but are not essential for in vitro receptor binding or growth stimulation of cells in culture.¹⁰⁴ Considerable heterogeneity in the glycosylation of circulating endogenous EPO results in multiple isoforms with different numbers of sialic acid residues. Isoforms with higher sialic content have a more prolonged half-life in the circulation and induce greater stimulation of erythropoiesis, despite their lower affinity for the EPO receptor.¹⁰⁵ A bioengineered, hyperglycosylated recombinant EPO, called darbepoetin, carries two additional N-linked carbohydrate chains with up to 22 sialic acid residues; endogenous EPO has a maximum of 14.¹⁰⁶ Despite an approximately fivefold lower affinity for the EPO-R, darbepoetin exhibits a half-life in the circulation approximately three times longer than that of EPO.^{107,108} Gross and Lodish¹⁰⁵ have developed an

in vitro model accounting for the prolonged bioactivity of darbepoetin. They found that darbepoetin and EPO have similar signaling properties and rates of internalization when bound to the EPO-R, but that EPO dissociates at a much slower rate from the EPO-R than darbepoetin, so more EPO is internalized and degraded.

EPO is produced primarily by the liver in the fetal period; after birth, the kidneys become the major source of production.^{72,95,109-111} Clearance of circulating EPO occurs by mechanisms that have not yet been fully elucidated. The liver, kidneys, and bone marrow have all been studied as possible sources of EPO elimination. A small fraction of endogenous or exogenous EPO appears to be cleared by filtration into the urine.¹¹² EPO degradation products can be found in urine, but the location and mechanisms responsible for this degradation are not known.¹⁰⁰

An important determinant of the fate of the circulating EPO is its binding to the EPO-R on erythroid cells¹¹³; the relative abundance of erythroid progenitors and precursors displaying EPO-Rs modulates serum EPO concentrations.¹¹⁴ The EPO-R is a 55-kDa transmembrane protein of the cytokine receptor superfamily,¹¹⁵⁻¹²⁰ displayed on erythroid progenitors from the colony-forming unit erythroid

(CFU-E) stage to late basophilic erythroblasts.¹¹⁵ The number of surface receptors has been estimated to be around 1000/cell. Upon binding of EPO to EPO-R, the sequence of signaling events are homodimerization of the receptor, which also undergoes a conformational change; generation of the intracellular signal; and clathrin-mediated endocytosis and proteolysis of the ligand-receptor complex, which ultimately determines the clearance of EPO from the circulation.^{105,121-126}

The EPO-R signal transduction pathway depends on the activation of Janus tyrosine kinase 2 (JAK2), which is physically associated with cytoplasmic portion of receptor, and becomes phosphorylated when the conformation of the receptor is changed by the binding of EPO to the external moiety.¹²⁷⁻¹³⁴ Activated JAK2 phosphorylates several of the eight tyrosine molecules of the cytoplasmic side of the EPO-R, exposing SH2 (src homology 2) binding sites for key signaling proteins,^{135,136} which result in a cascade of signal transduction activating multiple pathways including Ras/MAP kinase, JNK/p38 MAP kinase, JAK/STAT, the p85 regulatory subunit of phosphoinositide 3-kinase (PI3K), and protein kinase B (AKT).¹³⁷⁻¹⁴² Both JAK2 and the tyrosines on the cytoplasmic portion of the EPO-R appear to play a role in the internalization process.¹⁴³ Familial forms of polycythemia due to truncation mutations in the EPO-R with absence of key tyrosines such as Y429 and Y431 result in defective internalization of the EPO-R complex, prolong signal transduction, and increase EPO sensitivity.¹⁴³ The EPO-R endocytic machinery is critically dependent on a Cbl/p85/epsin-1 pathway, which ultimately leads to receptor downregulation.¹⁴⁴

After JAK2-mediated phosphorylation, STAT5 becomes activated and undergoes homodimerization; it translocates to the nucleus and activates EPO-inducible genes.¹⁴⁵ Transgenic mice lacking both STAT5a and STAT5b develop fetal anemia as increased apoptosis of erythroid progenitors decreases their survival.^{146,147}

The signaling induced by the binding of EPO to the EPO-R eventually results in an increased number of erythroid progenitors and precursors, in particular CFU-E.¹⁴⁸ Activation of parallel molecular pathways eventually suppress EPO-R signaling via tyrosine phosphatases, which dephosphorylate and inactivate JAK2; downregulate the EPO-R on the cell surface; and induce negative regulators such as CIS/SOCS (cytokine-inducible, SH2-containing protein-suppressor of cytokine signaling).^{138,149-156}

ERYTHROPOIETIC PROGENITOR/PRECURSOR CELLS AND EPO ACTION

An important aspect of the overall mechanism of action of EPO has been elucidated by Koury and Bondurant,¹⁴⁸ who demonstrated that EPO does not directly stimulate erythroid proliferation but rather prevents the programmed death (apoptosis) of the erythroid progenitors (Fig. 53.3).¹⁵⁷⁻¹⁵⁹ This EPO-dependent period is preceded by progenitors differentiating from multipotent hematopoietic stem cells (HSCs) to progenitor stages restricted to erythroid differentiation.¹⁶⁰ HSC differentiation is mediated by transcription factors GATA-1, friend of GATA-1 (FOG-1), and PU.1, with increased GATA-1 activity relative to PU.1 directing differentiation toward bipotent megakaryocyte-erythroid progenitors (MEPs).^{161,162} MEPs are directed

toward an erythroid fate by transcription factor KLF1 and toward a megakaryocyte fate by transcription factor FLI-1.¹⁶³ MEPs survive in response to either EPO and thrombopoietin (TPO), but their differentiation is determined by neither EPO nor TPO.¹⁶⁴ However, MEP fate can be directed toward megakaryocytes by low iron availability.¹⁶⁵

MEPs overlap with the most immature burst-forming units-erythroid (BFU-Es).¹⁶⁶ BFU-Es, defined by their generation of clustered erythroid colonies in culture, are the last stage of erythroid progenitors that circulate in the blood¹⁶⁷ and the first stage exclusively committed to the erythrocyte lineage.¹⁶⁸ Circulating BFU-Es lodge in the marrow via nonintegrin receptors for laminin,¹⁶⁹ where their descendants associate with marrow macrophages by an interaction involving macrophage ephrin-2 and erythroid cell receptors ephB4 and ephB6 that creates erythroblastic islands (EBIs).¹⁷⁰ EBIs are the marrow erythropoietic niches, each consisting of a central macrophage surrounded by differentiating erythroblasts.¹⁷¹

Within EBIs, the CFU-E stage of differentiation is defined by colony formation in culture and overlaps with the proerythroblast stage defined, like all erythroblast stages, by its microscopic appearance in stained needle-aspirated marrow smears. Modern flow cytometry techniques allow identification and isolation of murine^{147,172} and human¹⁷³ erythroid progenitors of BFU-E through proerythroblast stages using markers CD44, CD34, CD36, endoglin (CD105), transferrin receptor (CD71), and erythroid precursors of basophilic through orthochromatic erythroblasts using markers TER-119, glycophorin A, anion transporter (Band 3), and $\alpha 4$ integrin for the precursors. In addition to GATA1 and KLF1, the transcription factor complex of TAL1/SCL-LMO2-LDB1 is essential for all of these stages of erythroid differentiation.^{163,174}

In periods of physiologic hypoxic stress, BFU-Es and CFU-Es can expand their populations without differentiating further. In addition to EPO, factors mediating these expansions include stem cell factor (SCF/Kit-ligand), glucocorticoids, bone morphogenetic protein 4, and various adhesion molecules and their respective binding partners within EBIs.¹⁷⁵ However, the specific hypoxia-responsive factor is EPO, and specific erythropoietic stages that require EPO to prevent apoptosis are the CFU-E/proerythroblasts. CFU-E/proerythroblasts express the highest surface concentration of EPO-Rs.^{168,176,177} Without EPO, these cells are rapidly lost to programmed cell death.¹⁷⁸⁻¹⁸⁰ The finely graded survival responses of CFU-E/proerythroblasts to the hypoxia-induced, exponential increases in EPO are related to substantial heterogeneity in their responsiveness, likely related to the number of EPO-Rs, their functional status, or both.¹⁸¹ Some of this heterogeneity is due to a regulatory mechanism whereby EPO-dependent progenitors have variable surface displays of proapoptotic Fas (APO-1/CD95) and other erythroid progenitors/precursors constitutively produce or display Fas-ligand within the erythroblastic island.¹⁸²⁻¹⁸⁴ Fas activation in EPO-dependent progenitors promotes their apoptotic death by enhanced caspase-mediated cleavage of GATA-1,¹⁸⁵ whereas EPO decreases Fas expression.¹⁸⁴

EPO quantities are traditionally expressed in units, with 1 unit (U) representing the same erythropoietic effect in animals as 5-mmol cobalt chloride.⁹⁵ Steady-state

production of small amounts of EPO maintains the serum EPO concentration at approximately 10 to 30 U/L, enough to stimulate sufficient production of erythrocytes to replace those lost to senescence.^{69,95} When anemia or hypoxia is present, serum EPO concentrations increase rapidly, to as much as 10,000 U/L.⁷²⁻⁷⁴ Human studies have indicated a sustained increase in serum EPO concentrations after phlebotomy, with values remaining elevated for several weeks.^{186,187} In chronic anemia due to absence of erythroid progenitors, as with pure red cell aplasia (PRCA) and aplastic anemia, serum EPO values remain chronically elevated, as much as 1000-fold higher than normal.¹⁸⁸⁻¹⁹²

Koury and Bondurant and colleagues^{36,148,193} have incorporated these basic physiologic concepts into a model that explains how EPO regulates erythropoiesis in various pathologic conditions (see Fig. 53.3). In this model, the EPO-dependent phase of erythropoiesis encompasses three generations (from CFU-E through early erythroblasts), with each generation having a certain proportion of surviving cells and the remaining cells being lost by apoptosis. Owing to their reduced EPO responsiveness (or greater EPO dependence), most of the cells at the CFU-E stage succumb to apoptosis, so the erythropoietic production flow in a healthy individual arises from a minority of progenitors that have escaped apoptosis. When EPO concentrations increase in response to hypoxia, blood loss, or hemolysis, additional progenitors that would be lost to apoptosis under normal conditions now survive and, a few days later, result in increased absolute numbers of reticulocytes and ultimately of RBCs. If this response is sufficient to compensate for the decreased oxygen-carrying capacity of blood, EPO concentrations decline and erythropoiesis returns to normal rates. When EPO production is impaired, such as in CKD, a greater number of progenitors become apoptotic, EPO concentrations are insufficient to maintain an adequate pool of differentiating progenitors, and the resulting impaired reticulocyte production ultimately leads to anemia.

Basal and EPO-stimulated erythropoiesis require contact between erythroblasts and central macrophages in EBIs.^{193,194,171} Surface displayed or secreted products of central macrophages stimulate erythroid cell proliferation including SCF,^{195,196} ephrin-2,¹⁹⁷ and bone morphogenetic protein 4 (BMP4).¹⁹⁸ Conversely, negative regulators of erythroid cell growth also have been identified in EBIs including the inflammatory cytokines tumor necrosis factor- α (TNF- α), transforming growth factor- β (TGF- β), TNF-related apoptosis-inducing ligand (TRAIL), and receptor-binding cancer antigen of SiSo cells 1 (RCAS1).¹⁷¹ Interferon- γ (IFN- γ) increases Fas expression on EPO-dependent progenitors but does not affect the production of Fas-ligand in the erythroblastic island,¹⁸³ while EPO suppresses progenitor Fas expression.¹⁸⁴ Central macrophages play a role in the proliferation and final enucleation of erythroblasts, but they also supply the developing erythroblasts with ferritin, iron, and recycled DNA components from nuclei ingested after enucleation of orthochromatic erythroblasts in EBIs.¹⁹⁹⁻²⁰¹ During recovery from bleeding or hemolysis, central macrophages transfer mitochondria, sources of energy and sites of the first and last reactions of heme synthesis, to early-stage erythroblasts by a mechanism involving CD47.²⁰²

THE ROLES OF IRON, FOLATE, AND VITAMIN B₁₂ IN ERYTHROPOIESIS

In late erythropoietic stages, hemoglobin-synthesizing erythroblasts progressively decrease their size while condensing nuclear chromatin before the enucleation that forms the reticulocyte, which subsequently enters the circulation, and the nucleus-containing pyrenocyte, which is ingested by the EBI central macrophage.¹⁶⁰ Reduced cell size is accomplished by rapid cell division with a shortened G1 phase of cell cycle induced by direct contact with a central macrophage¹⁹³ and regulated by cyclin D3.²⁰³ Three nutrients are crucial for late-stage erythropoiesis: iron for hemoglobin production and folate and vitamin B₁₂ for rapid cell division.²⁰⁴

Reduced folate, in the form of tetrahydrofolate (THF), is a cofactor in the one-carbon metabolism of three steps of nucleoside synthesis required for DNA and RNA syntheses: Two carbons are provided for synthesis of purine ribonucleosides and deoxyribonucleosides, and one carbon is provided in the synthesis of the pyrimidine deoxynucleoside, thymidylate.²⁰⁴ Vitamin B₁₂ is indirectly involved in nucleotide synthesis as the cofactor required to recover THF that has accumulated in the methyl-THF form, as vitamin B₁₂ transfers the methyl group from methyl-THF to homocysteine in the synthesis of methionine.^{205,206}

Folate and/or cobalamin deficiencies have their greatest effects on erythroid cells as they are in transition from the EPO-dependent stages to the Hgb synthesis stages.^{207,208} Inadequate deoxynucleosides impairing DNA synthesis lead to anemia by erythroid progenitor/precursor apoptosis in the S phase of the cell cycle.²⁰⁸ However, folate-deficient erythroblasts that can survive in the prolonged S-phase continue to synthesize protein and become larger than normal, thus being termed megaloblastic.²⁰⁷ These megaloblastic erythroblasts in turn give rise to larger-than-normal reticulocytes and RBCs (i.e., macrocytic anemia). The inefficient erythropoiesis of megaloblastic anemias is characterized by reduced numbers of reticulocytes, increased serum bilirubin and lactic dehydrogenase (LDH), and accelerated iron turnover.^{204,205}

Intracellular availabilities of iron, heme, and globin chains have to be perfectly matched because an imbalance of these constituents is toxic for the erythroblast.^{209,210} About 1 mg of iron is lost daily by shedding of gut and skin cells, and this loss is balanced by absorption of 1 mg of inorganic iron daily through the duodenum.²⁰⁴ The large majority of iron used for erythropoiesis is recycled from senescent RBCs that have been phagocytosed by macrophages. As each milliliter of blood contains about 0.5 mg of iron, small, long-term blood losses eventually deplete body iron stores with development of iron-deficient erythropoiesis and anemia.²⁰⁴ Iron is crucial for basic cellular processes such as DNA synthesis and mitochondrial function, and iron deficiency suppresses erythropoiesis at multiple stages leading to decreased reticulocyte production and anemia, while sparing iron for other uses in other cell types. Iron deficiency directs MEPs toward a megakaryocytic differentiation.¹⁶⁵ EPO concentrations are lower in iron deficiency anemia than many other anemias because EPO induction in renal interstitial fibroblasts is diminished by suppression of HIF-2 α synthesis, due to IRP-1 binding and inhibition of HIF-2 α mRNA translation.⁸⁵⁻⁸⁷ Lastly, iron

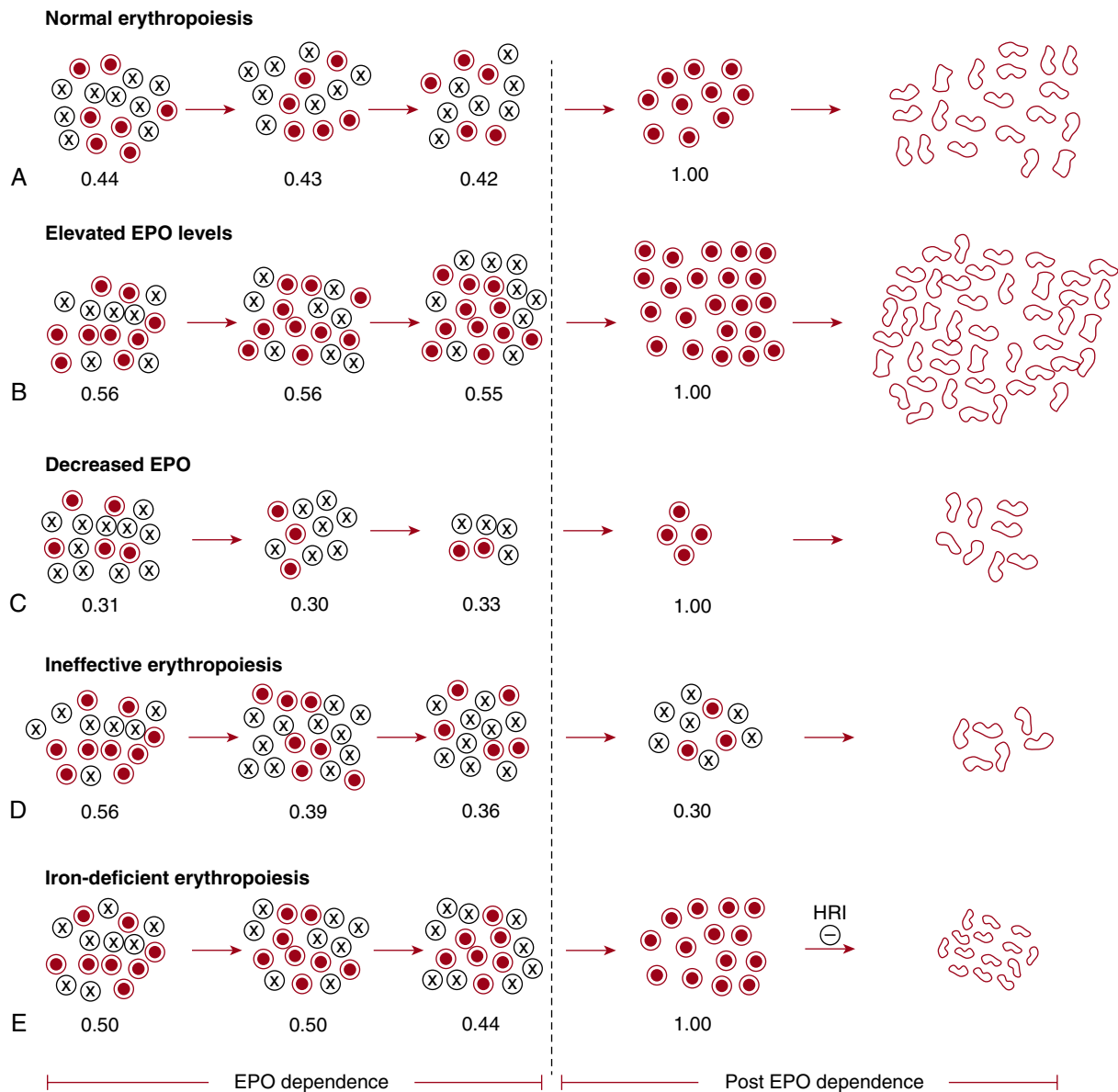


Fig. 53.3 Model of erythropoiesis based on suppression of apoptosis by EPO and heterogeneity in erythropoietin (EPO) dependence among erythroid cells. From CFU-E through early basophilic erythroblast stages, erythroid progenitor/precursor cells depend on EPO for survival. The EPO-dependent period is left of the *dotted line* and encompasses three generations and two cell divisions. Each division is represented by an *arrow*. In the post-EPO-dependent period, to the right of the *dotted line*, two cell divisions occur. Surviving cells in each generation are shown as *circles* containing large *red dots* representing intact nuclei. Cells succumbing to apoptosis are shown as *circles* containing *Xs*. The proportion of the total cells that survive is shown below each generation. The number of surviving cells in a generation results in twice that number of total cells in the subsequent generation. The final populations of cells shown on the right represent the anucleate, irregular reticulocytes. (A) Normal erythropoiesis, with average survival rates of 43% in the EPO-dependent generations, produces 200–250 billion reticulocytes daily. (B) Elevated EPO levels as found after acute blood loss or hemolysis increase survival rates to 56% in the EPO-dependent generation, and reticulocyte production rate more than doubles. (C) Decreased EPO levels, as found in renal failure, decrease survival rates to 32% in the EPO-dependent generation, and reticulocyte production is less than one-half of normal. (D) Ineffective erythropoiesis increases rates of apoptosis due to a pathologic process such as folate or vitamin B₁₂ deficiency or β -thalassemia. High EPO levels in response to decreased erythrocyte production expand surviving cells in the early EPO-dependent generations, but the increased rates of apoptosis in the late EPO-dependent and post-EPO-dependent stages decrease daily reticulocyte production to less than one-third of normal. (E) Iron-deficient erythropoiesis with moderately elevated EPO activity for the degree of anemia slightly increases the survival rate during the EPO-dependent period, but influx from pre-EPO dependent stages is decreased as fewer MEPs are committed to erythropoiesis. In the post-EPO-dependent period, when hemoglobin is synthesized, heme-regulated inhibitor (*HRI*) prevents apoptosis by inhibiting general protein synthesis while enhancing the production of the mediator of stress erythropoiesis, the transcription factor ATF4. The inhibited protein synthesis decreases reticulocyte production rate, size, and hemoglobin content. (From Koury MJ, Blanc L. Red blood cell production and kinetics. In: Simon TL, Gehrie E, McCullough J, Roback JD, Snyder EL, eds. *Rossi's Principles of Transfusion Medicine*. 6th ed. Hoboken, NJ: Wiley-Blackwell; 2022:133–142.)

is preferentially exported from hemoglobin-synthesizing erythroid cells and duodenal enterocytes in iron deficiency because these two cells, which supply other cell types with iron during iron deficiency, have transcripts for the universal cellular iron exporter, ferroportin that lack an IRE. In iron deficiency, erythroid cells and duodenal enterocytes can export iron,^{211,212} while the other cell types with ferroportin transcripts containing IREs have suppressed ferroportin expression, allowing conservation of intracellular iron.

Within the erythroblast, deficient iron results in decreased heme synthesis that, in turn, suppresses hemoglobin production. Decreased heme inhibits β -globin transcription by restricting heme's allosteric derepression of the Bach1 transcription inhibitor, but decreased intracellular heme causes broader, profound suppression of hemoglobin production by decreased translation of mRNAs encoding α -globin, β -globin, and many other erythroblast proteins due to a reduction in heme's allosteric effects on eIF2 α kinase (heme-regulated inhibitor of translation/HRI).^{210,213} When iron deficiency's negative effects on erythroid precursor numbers due to suppressed differentiation and relatively decreased EPO production are combined with diminished production of hemoglobin and other proteins in erythroblasts, the result is production of fewer and smaller RBCs containing less hemoglobin (i.e., microcytic anemia).^{210,213} In macrophages, HRI acts as a positive modulator of cytokine and hepcidin production, thus affecting both EPO signaling and iron metabolism.^{214–216}

PATHOBIOLOGY OF THE ANEMIA OF CHRONIC KIDNEY DISEASE

Anemia in CKD can develop because of any of the diseases or deficiencies that may affect individuals without kidney disease, such as iron deficiency, vitamin B₁₂,^{217,218} or folic acid deficiency,²¹⁷ and chronic blood loss.²¹⁹ However, the form of anemia most common in CKD is a normocytic, normochromic, or slightly hypochromic²²⁰ anemia, with insufficient production of erythrocytes.^{221–224} The cause of low Hgb is multifactorial, with contributors such as volume overload,²²⁵ relative EPO deficiency, iron deficiency, blood loss, hemolysis, chronic inflammation,²²⁶ drugs such as nonsteroidal anti-inflammatory drugs (NSAIDs), and other factors, which may include circulating cytokine inhibitors of erythropoiesis.^{224,227–229} The preponderance of evidence has demonstrated that EPO deficiency is the major cause of anemia in patients with CKD.^{230–233} Ultimately, the best evidence for the primacy of EPO deficiency in the pathogenesis of CKD-related anemia has been the consistent success of treatment with rhEPO or its derivatives. Other contributing causes to anemia should be considered if the severity of anemia is much greater than expected, if higher than usual doses of rhEPO are needed, and in the presence of leukopenia or thrombocytopenia.

ERYTHROPOIETIN PRODUCTION AND KIDNEY DISEASE

In normal persons, serum EPO concentrations rise exponentially and inversely in response to reductions in red cell mass.²³⁴ In people with CKD, EPO concentrations are inappropriately low for the degree of anemia but may still be similar to or even higher than those in

normal, nonanemic subjects.^{235–239} The adequacy of EPO production in response to anemia appears to decline in rough proportion to the degree of reduction in nephron mass.^{234,240,241} At all stages of CKD, serum EPO values were found to be higher than in nonanemic normal subjects, but the relationship between Hgb and serum EPO depends on the severity of the CKD.²⁴² Among patients with mild-to-moderate CKD, the correlation was inverse, with lower Hgb concentrations being associated with higher than normal serum EPO concentrations, but not as high as in similarly anemic patients without CKD or other chronic illnesses associated with inflammation. However, among patients with creatinine clearances below 40 mL/min, mean serum EPO concentrations were severely depressed and uncorrelated with the degree of anemia but directly correlated with creatinine clearance, indicating a parallel loss of renal excretory and EPO-producing capacities.²⁴² Fehr and associates²⁴³ studied 395 patients undergoing coronary angiography, 84% of whom had reduced creatinine clearance values, and found that serum EPO concentrations were higher in patients with lower Hgb, except when creatinine clearance was below 40 mL/min. Similarly, Artunc and Risler²⁴⁴ observed that the correlation between hemoglobin and EPO concentrations weakens progressively with advancing stages of CKD, becoming negligible in stages G4 and G5.

Why EPO production by diseased kidneys is inadequately low remains incompletely understood. One concept proposes that EPO production is reduced because of the transformation of peritubular fibroblasts with EPO-producing capacity into myofibroblasts that cannot produce EPO.^{245,246} On the other hand, stabilization of HIF with inhibitors of prolyl hydroxylases results in significant EPO secretion from renal and extrarenal tissues, even in patients undergoing dialysis.²⁴⁷ Thus, a disturbed oxygen-sensing mechanism rather than destroyed production capacity for EPO appears to be the primary cause of CKD-related anemia. In fact, despite the severely diminished EPO response with advanced CKD, some degree of sustained feedback remains. In the 6 months before starting dialysis, Radtke and colleagues²⁴² found that as anemia worsened, serum EPO concentrations increased, and in the 6 months after the start of dialysis, the opposite occurred. This continued response to anemia in patients with advanced CKD was also demonstrated by Walle and colleagues,²⁴⁸ who found that serum EPO values increased after hemorrhage and declined after blood transfusion in patients on dialysis. Hypoxia can increase EPO production significantly in anemic patients with CKD^{249,250} and, among patients with CKD and similar degrees of anemia, anephric patients had lower plasma EPO levels than those with kidneys.²³⁴ Consistent with a residual capacity for endogenous EPO secretion in patients with CKD, a large analysis revealed that with higher altitude—and thus lower blood oxygen content for any given Hgb concentration—higher achieved Hgb values were observed in patients on maintenance dialysis, even though lower doses of ESAs were used.³¹

Significantly diminished EPO production in response to anemia when creatinine clearance is below 40 mL/min is consistent with clinically relevant anemia becoming common only with moderate to severe CKD (see earlier discussion and Fig. 53.1).^{13,252}

SHORTENED RED CELL SURVIVAL

Using various labeling methods, reduced red cell survival has been documented in anemic patients with CKD.²⁵³⁻²⁵⁵ Proteinuria and nephrotic syndrome are also associated with shortened RBC lifespan.²⁵⁶ Indirect measurement of endogenous carbon monoxide, which is produced along with iron by heme oxygenase-1 catabolism of heme to biliverdin, confirmed that RBC lifespan progressively shortens as kidney function declines.²⁵⁷ The degree to which decreased RBC lifespan contributes to the CKD-related anemia is uncertain, but premature removal from the circulation of stressed RBCs (eryptosis) appears to play a significant role.²⁵⁸⁻²⁶⁰ Several abnormalities have been described in uremic erythrocytes, which may result in their increased premature destruction. An abnormal externalization of phosphatidylserine (PS), a phospholipid normally present only on the inside of the RBC membrane, has been associated with increased erythrophagocytosis and anemia in CKD.²⁶¹ Uremic red cells have been reported to become more fragile in response to osmotic stimuli,²⁶² although this finding was not confirmed in pediatric patients undergoing peritoneal dialysis.²⁶³ The rheologic properties of uremic erythrocytes are altered owing to changes in RBC shape and decreased deformability.²⁶⁴ Uremic erythrocytes may not be able to mount an effective response to oxidative stress,²⁶⁵⁻²⁶⁷ possibly because of glutathione deficiency,²⁶⁸ and may benefit from the antioxidant effects of vitamin E bound to dialysis membranes.^{267,269} Carnitine deficiency may also contribute to the reduced survival of uremic erythrocytes.^{270,271} An abnormal deposition of complement onto erythrocytes in CKD can play a role in their premature removal from the circulation.²⁷²

BLOOD LOSS

Excessive bleeding has long been recognized as a common and significant complication of CKD.²⁷³ The coagulopathy of CKD, discussed in the final section of this chapter, likely plays a major role in occult blood loss via gastrointestinal bleeding of patients with CKD.²⁷⁴ In addition, blood loss associated with dialysis procedures and laboratory studies is also significant. Fifty years ago annual blood loss due to hemodialysis was estimated to be 1 to 3 L.²⁷⁵ Subsequent improvements in hemodialysis techniques and clinical laboratory methodologies reduced this loss considerably, with later estimates of the blood lost in the whole extracorporeal circuit for each dialysis session varying from a range of 1.5 to 1.8 mL²⁷⁶ in one study to a median of 2.9 mL in another.²⁷⁷ Each milliliter of blood contains approximately 0.5 mg of iron, so an important consequence of blood loss is the loss of iron and the development of iron deficiency (see next).

METABOLIC AND CYTOKINE INHIBITION OF ERYTHROPOIESIS

A variety of uremic toxins have been identified in CKD^{278,279} including some with hematologic effects, such as quinolinic acid²⁸⁰ and *N*-acetyl-seryl-aspartyl-lysyl-proline (AcSDKP),²⁸¹ but they do not appear to play a significant role in the anemia of CKD. Responses to ESAs can be improved with dialysis, and EPO doses used to treat patients with anemia in CKD are much higher than endogenous EPO produced by normal individuals, indicating reduced responsiveness.

However, no significant reduction in ESA doses used to manage anemia were observed for frequently hemodialyzed patients (six times/week) compared with those receiving conventional hemodialysis (three times/week).²⁸²

Cytokines produced by immune cells that cause chronic inflammation are major contributors to anemia in CKD, and their effects in patients without CKD but with the anemia of chronic disease (ACD) provide insights into the roles of cytokines in the anemia of CKD.^{283,284} In early-stage erythroid progenitors, TNF- α ,²⁸⁵⁻²⁸⁷ IL-1,^{287,288} and IFN- γ ²⁸⁹ have been shown to promote differentiation toward monocytes and granulocytes at the expense of erythroid differentiation. In these progenitors, GATA-1 promotes erythropoiesis and PU.1 promotes myelopoiesis, thereby competing in determining differentiation fate.^{290,291} In mice, INF- γ induces PU.1, inhibits erythropoiesis, and results in anemia.²⁸⁹ The negative effects of TNF- α and IL-1 on erythroid progenitor survival and differentiation are indirect in that the negative effects of TNF- α require INF- γ ,²⁹² while those of IL-1 require IFN- β .²⁹³ Two other inflammatory cytokines that contribute to ACD are IL-33²⁹⁴ and IL-22,²⁹⁵ both of which bind their respective receptors on erythroid progenitors, thereby inhibiting differentiation of those progenitors. Itaconate, a metabolite produced by macrophages in chronic inflammation, is a competitive inhibitor of erythroid-specific aminolevulinate synthetase-2 (ALAS-2), the rate-limiting enzyme in porphyrin synthesis.²⁹⁶ Itaconate limiting production of protoporphyrin in erythroid cells contributes to suppressed heme synthesis and anemia.

In ACD, as in the anemia of CKD, erythropoiesis is suppressed by inadequate EPO leading to the apoptotic loss of EPO-dependent progenitors. Like patients with CKD, those without CKD but with ACD due to rheumatoid arthritis have inadequate plasma EPO levels for the degree of anemia,^{297,298} although EPO levels in ACD are not as suppressed as in severe CKD. Inflammatory cytokines TNF- α , IL-1, or IL-6 inhibit hypoxia-induced EPO production in isolated kidneys.²⁹⁹ In the marrow, EPO-dependent progenitors display surface receptors for stem cell factor and EPO, and both cytokines promote survival, but INF- γ downregulates surface expression of both receptors³⁰⁰ while it also increases Fas expression that induces erythroblast apoptosis.¹⁸² In addition to decreased EPO and EPO-R, ACD also induces increased concentrations of high mobility group box-1 (HMGB1) protein that disrupts EPO-EPO-R interaction and decreases EPO-R signaling in EPO-dependent progenitors.²¹⁵

IRON METABOLISM, HEPCIDIN, AND INFLAMMATION

A major component of ACD and the anemia of CKD is decreased iron availability secondary to generalized cellular retention of inorganic iron that is mediated by elevated plasma hepcidin.^{214,301} Plasma membrane expression of ferroportin, the only known exporter of inorganic iron,³⁰²⁻³⁰⁴ is regulated by IRP1 in all cells except erythroid cells and duodenal enterocytes, where ferroportin transcripts lack an IRE.^{211,212} However, the major regulator of iron availability is the hepatic hormone hepcidin,³⁰⁶ which binds ferroportin, thereby blocking its export function and inducing internalization and degradation.³⁰⁷⁻³⁰⁹ Plasma hepcidin increases in response to inflammatory cytokines,³¹⁰ mainly

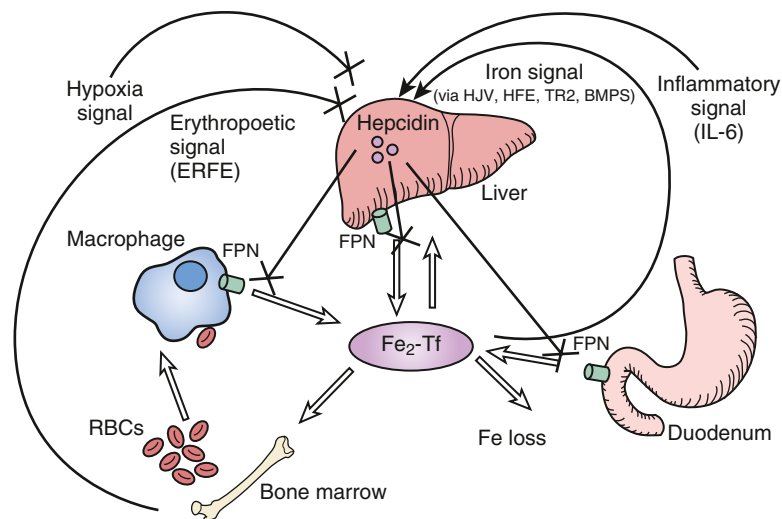


Fig. 53.4 Hepcidin is a central regulator of systemic iron homeostasis. Serum iron concentrations are determined by the balance of iron entry from intestinal absorption, macrophage iron recycling, and mobilization of hepatocyte stores versus iron utilization, primarily by erythroid cells in the bone marrow. A peptide hormone secreted by the liver, hepcidin controls iron release into the plasma by downregulating cell surface expression of the iron export protein ferroportin (FPN) on all cell types, including absorptive enterocytes, macrophages, and hepatocytes. A complex interaction of transferrin-bound iron, transferrin receptor 2 (*TFR2*), hemojuvelin (*HJV*), hereditary hemochromatosis (*HFE*), bone morphogenetic proteins (*BMPs*), and BMP receptors regulate hepcidin production. Hepcidin production is stimulated by iron as a negative feedback loop to maintain steady-state iron concentrations. Hepcidin production is inhibited by erythroferrone (*ERFE*), a hormone produced by erythroid cells during expanding erythropoiesis that binds and sequesters BMPs. Hepcidin production also is stimulated by inflammation, mainly by the inflammatory cytokine interleukin-6 (*IL-6*), thereby causing the hypoferremia of anemia of chronic disease. *Fe₂-Tf*, Transferrin-bound iron; *RBC*, red blood cell. (Modified from Babitt JL, Lin HY. Molecular mechanisms of hepcidin regulation: implications for the anemia of CKD. *Am J Kidney Dis*. 2010;55:726–741.)

via IL-6 and its receptor's signaling by the Jak2/Stat3 pathway,³¹¹ although IL-1 signaling via the SMAD pathway has also been reported.^{312,313}

Hepcidin also has a regulatory role in iron deficiency and overload. In mice, hepcidin deficiency increases iron absorption and accumulation in the liver,³¹⁴ similar to patients with mutant hepcidin (*Hamp*) hemochromatosis,³¹⁵ while hepcidin overexpression in mice leads to severe iron deficiency.³¹⁶ This iron-hepcidin regulatory pathway involves a complex interaction involving transferrin-bound iron, transferrin receptors 1 and 2, hemojuvelin, hereditary hemochromatosis (*HFE*), and bone morphogenetic proteins (*BMPs*) 2 and 6 produced by hepatic endothelial cells and respective BMP receptors and the SMAD signaling pathway in hepatocytes.^{214,301} This BMP pathway for hepcidin production is regulated by plasma erythroferrone (*ERFE*),^{317,318} a hormone produced by erythroid cells, and a hepatocyte membrane protein, transmembrane serine protease 6 (*TMPS6*),^{319,320} which is increased in hepatocytes in low iron states.³²¹ *ERFE* appears to bind and sequester *BMP6* and its heterodimers with *BMP2*,³²² thereby disrupting its receptor binding, while *TMPS6* binds and inactivates the hemojuvelin component of the BMP receptor complex.³²¹

The identification of hepcidin as a key regulator of iron homeostasis and its role as a response to inflammation has redefined our understanding of iron availability in CKD (Fig. 53.4).^{214,301,323-330} In anemic patients with CKD, hepcidin is the major regulator for both iron deficiency and restricted iron availability, described in the clinical setting as functional iron deficiency, and its role is now recognized in the maintenance of adequate iron for erythropoiesis

while avoiding the negative consequences of iron overload.^{273,331-333} In the absence of chronic inflammation, blood loss leads to a reduction of serum ferritin and serum iron values with a progressive increase in the desaturation of transferrin, below the 16% threshold that guarantees an adequate supply of iron to the erythroid marrow.³³⁴ Compared with normal individuals, both impaired^{335,336} and unimpaired iron absorption^{337,338} has been reported in patients on maintenance dialysis. However, initiation of EPO therapy increases iron absorption over baseline.^{337,338} In this regard, anemia, EPO administration, and hypoxia increase iron absorption and mobilization by decreasing hepcidin production,^{339,340} although hepcidin is not a direct target gene of *HIF*.³⁴¹

Although standardized assays for hepcidin for clinical use are not yet available, several studies have measured hepcidin concentrations in CKD but with potentially conflicting results.³⁴² These studies must be considered with two caveats. First, hepcidin concentrations are reduced in anemic persons with normal kidney function; thus a normal hepcidin concentration in CKD may still be inappropriately high for the degree of anemia. Second, hepcidin concentrations in healthy persons are reduced by 70% to 75% at 24 hours after EPO administration.³⁴³

Residual kidney function, iron stores, erythropoiesis status, and inflammation all seem to be related to the hepcidin concentrations observed in CKD.³⁴⁴ Hepcidin is cleared from plasma by the kidneys,²¹⁴ and progression or severity of anemia in patients with ND-CKD appears to be associated with higher serum hepcidin concentrations.^{345,346} Baseline values of, or changes over time in, serum hepcidin did not predict

a response to iron therapy in patients with ND-CKD.³⁴⁷ In patients on dialysis, hepcidin concentrations in one study were elevated but they were not correlated with IL-6 concentrations or responsiveness to treatment, although they decreased after initiation of EPO therapy.³⁴⁸ Elevated serum hepcidin concentrations in patients with CKD were associated with elevated ferritin and/or CRP concentrations and with stage 5 CKD.³⁴⁹ Serum hepcidin concentrations were reduced following dialysis treatment.^{350,351} Hepcidin concentrations also may vary in patients on hemodialysis who have *HFE* gene mutations, which reduce hepcidin production.³⁵² Other studies have reported normal concentrations of hepcidin in CKD but not addressed the caveats mentioned previously.^{350,353}

The reductions in hepcidin in CKD with stimulated erythropoiesis suggest erythroid cell production of ERFE is playing a role because ERFE is an EPO-driven negative regulator of hepcidin production.³¹⁷ In patients with CKD, endogenous serum EPO values or weekly therapeutic EPO doses were directly correlated with serum ERFE values,³⁵⁴ but ERFE values did not appear to influence serum hepcidin values, suggesting other regulatory mechanisms play a role in determining hepcidin values in CKD. Conversely, in another study of patients receiving maintenance hemodialysis, treatment with ESAs was associated with increased concentrations of ERFE and reduced concentrations of serum hepcidin.³⁵⁵ With the provision of iron supplementation with EPO administration, hepcidin-mediated inhibition of iron export from hepatic cells that store iron and reticuloendothelial cells that recycle iron has become the focus of decreased plasma iron in CKD-related anemia.^{283,331-333} Increased plasma transferrin concentrations that accompany iron deficiency anemia do not occur in ACD and anemia of CKD, so the percentage of serum transferrin saturation (serum iron/total iron binding capacity) is normal or increased.^{214,283,310} Conversely, hepcidin-mediated accumulation of iron in hepatocytes and reticuloendothelial cells leads to increased intracellular iron stored in ferritin, small amounts of which are released and detected as normal or increased serum ferritin concentrations as compared with the decreased serum ferritin concentrations of iron deficiency.^{214,310} Ferroportin-mediated export of iron that occurs in iron-deficient erythroblasts is mitigated in erythroblasts of ACD and anemia of CKD because elevated plasma hepcidin downregulates their expression of ferroportin.²¹² Therefore erythroblasts in CKD-related anemia demonstrate an attenuated reduction in intracellular iron as compared to their iron-deficient counterparts. The relatively increased intracellular iron in erythroblasts of CKD-related anemia compared with iron-deficient erythroblasts results in HRI-mediated production of slightly larger RBCs containing slightly more hemoglobin.^{212,356,357} Additionally, the shortened RBC lifespan in CKD-related anemia (described earlier) results in the average age of the RBC population in being younger and, therefore, larger.³⁵⁸ Thus despite reduced iron availability due to inflammation in CKD-related anemia, the combined effects of high plasma hepcidin and shortened RBC lifespans result in the characteristic normocytic anemia of CKD.

FOLIC ACID, VITAMIN D, AND ZINC DEFICIENCY

A net loss of folate is associated with dialysis, although the deficit is typically compensated by a normal diet and/or routine supplementation of water-soluble vitamins. Folate status is best assessed by measuring RBC folate because the plasma assay is affected by recent dietary intake and overestimates the true prevalence of folate deficiency.³⁵⁹ Changes in RBC parameters (e.g., increases in mean corpuscular cell volume [MCV] and mean corpuscular cell hemoglobin [MCH] from baseline) are helpful in identifying folate deficiency in patients on dialysis.^{360,361}

Vitamin D deficiency is an independent predictor of anemia in early stages of CKD,^{362,363} but controlled clinical trials of vitamin D supplementation in patients with kidney failure,^{364,365} pediatric patients with ND-CKD,³⁶⁶ and hemodialyzed patients³⁶⁷ failed to demonstrate any benefit of vitamin D related to anemia or its management. One study even suggested that among patients receiving hemodialysis, cholecalciferol supplementation increases serum hepcidin-25 levels in the short term and may increase erythropoietin resistance in the longer term.³⁶⁷

Low plasma zinc (Zn) concentration has been reported, with variable incidences in patients undergoing hemodialysis.³⁶⁹⁻³⁷¹ A Zn supplementation trial has shown measurable improvements in Hgb concentrations.³⁷² However, Zn supplementation, even within the serum normal range, can cause Zn toxicity, anemia, and pancytopenia by inducing copper deficiency.^{373,374}

ALUMINUM OVERLOAD

Formerly, aluminum was commonly used in patients on dialysis for its effects as a potent intestinal phosphate binder. Although calcium-containing and non-calcium-containing phosphate binders have largely supplanted aluminum, the effects of aluminum toxicity on hematopoiesis are of historic interest. Parenteral aluminum exposure, either via dialysate contamination³⁷⁵ or another route,³⁷⁶ is still observed. The erythropoietic effects of aluminum toxicity are characterized by altered iron metabolism,³⁷⁷ direct inhibition of erythropoiesis,^{378,379} and disruption of red cell membrane function and rheology.³⁸⁰⁻³⁸² In dialyzed patients, the most notable hematologic effect of aluminum overload is microcytic anemia,^{383,384} which improves with the use of deionized water to reduce the aluminum content of the dialysate³⁸⁵ or chelation therapy with desferrioxamine.³⁸⁶ EPO responsiveness is reduced in patients undergoing dialysis who have higher serum aluminum concentrations at baseline or after a deferoxamine challenge³⁸⁷ and can be restored with deferoxamine treatment.³⁸⁸ Improvement of anemia has also been shown in patients with the use of chelation therapy with deferoxamine, even in the absence of overt aluminum toxicity.^{389,390} Interestingly, HIF destabilization requires iron as a cofactor, and iron chelation with deferoxamine can induce HIF, providing a possible alternative explanation for the correction of anemia.

HORMONES AND MARROW FIBROSIS

Intact fibroblast growth factor 23 (FGF23) has been shown to be an important determinant of anemia and iron deficiency in mouse models of kidney failure.³⁹¹ Serum FGF23 concentrations were inversely corrected with Hb concentrations in 3869 patients with mild to severe CKD,³⁹²

with high FGF23 values being associated with a risk of developing anemia in a cohort of 1164 nonanemic patients³⁹³ and a reduction in serum concentrations after IV iron therapy.³⁹⁴ However, serum FGF23 concentrations showed no association with anemia in a cohort of 282 patients on hemodialysis.³⁹⁵

The inhibitory effects of parathyroid hormone (PTH) on erythropoiesis are primarily indirect and a consequence of myelofibrosis.³⁹⁶ Secondary hyperparathyroidism is associated with diminished responsiveness to EPO.³⁹⁷ Moreover, PTH values were identified as effect modifiers of the erythropoietic response to EPO in adult patients on hemodialysis.³⁹⁸ In 979 patients with ND-CKD, a significant inverse association has been reported between PTH and hemoglobin.³⁹⁹ However, in pediatric patients, no association was found between serum intact PTH and Hgb concentrations.⁴⁰⁰ Racial differences have been reported for serum PTH concentrations, with higher values in Black persons relative to White persons.⁴⁰¹ An increase in endogenous EPO and correction of anemia have been reported after subtotal parathyroidectomy.^{402,403}

DRUGS

Use of renin-angiotensin-aldosterone system (RAAS) inhibitors may induce or worsen anemia⁴⁰⁴ for several reasons. Angiotensin II has direct facilitating effects on erythroid progenitor cells, which are inhibited by these medications.⁴⁰⁵ AcSDKP, an endogenous inhibitor of erythropoiesis, accumulates in patients treated with angiotensin-converting enzyme (ACE) inhibitors.²⁸¹ Endogenous EPO production may also be reduced through the hemodynamic effects of angiotensin II inhibition. Because angiotensin II leads to preferential constriction of efferent glomerular arterioles, it increases the ratio of filtered sodium—the main determinant of renal oxygen consumption—to peritubular blood flow and thus oxygen supply, thereby presumably lowering peritubular oxygen tension. RAAS inhibitors reverse these effects and therefore have the potential to mitigate renal hypoxia and the signal for EPO production.⁴⁰⁶ It has also been postulated that RAAS inhibitors may promote anemia and EPO resistance via a reduction in testosterone serum concentrations in men younger than 60 years.⁴⁰⁷ Myelosuppressive effects of immunosuppression may further contribute to anemia, especially in the posttransplantation setting.⁴⁰⁸⁻⁴¹⁰

ASSOCIATION OF ANEMIA WITH ADVERSE OUTCOMES

Anemia is associated with reduced health-related quality of life (HRQOL)^{411,412} and prognosis of patients with CKD. A large number of observational studies have consistently shown that even modest reductions in Hgb concentrations are associated with adverse outcomes. This applies to mortality in patients on dialysis,⁴¹³ patients with ND-CKD,⁴¹⁴ and individuals in the general population⁴¹⁵ or with other complex chronic diseases, such as heart failure.⁴¹⁶ A large study of 159,720 patients undergoing hemodialysis and receiving rhEPO therapy showed that the duration of anemia, rather than the Hgb concentration per se, is the most powerful predictor of short-term mortality, with Hgb concentrations <11 g/dL for 3 months or longer

being associated with an increased risk of death.^{417,418} However, there is no agreement on how best to study Hgb variability effects on mortality, with various methods having been applied to describe Hgb variability⁴¹⁹ and with significant confounding attributable to variations in Hgb concentrations among dialysis centers⁴²⁰ and the effects of ESA dosing and iron therapy.⁴²¹ Hb concentrations below 10 g/dL and above 12 g/dL before kidney failure treated with replacement therapy (KFRT) have been associated with higher all-cause mortality and cardiovascular mortality, whereas Hb concentrations below 10 g/dL were associated with higher hospitalization rates in a study of more than 31,000 U.S. veterans.⁴²²

A systematic review has supported the notion that Hgb concentrations below a formerly established reference range (Hgb 9–10 g/dL in some studies and Hgb 11–12 g/dL in others) are generally associated with all-cause mortality in patients on dialysis.⁴²³ Similar findings have been reported in pediatric patients on peritoneal dialysis.⁴²⁴ In ND-CKD patients, the severity of anemia is associated with the rate of decline in kidney function,⁴²⁵ consistent with the concept that anemia may aggravate intrarenal hypoxia.⁴²⁶⁻⁴²⁸ Moreover, anemia was also found to be a strong risk factor for the development of left ventricular hypertrophy,^{429,430} an established surrogate for mortality and cardiovascular events. An increase in cardiac output, as part of the compensatory mechanisms that maintain oxygen delivery in anemia, has been considered as a possible reason for the link between anemia and cardiac geometry.⁴²⁹

In a relatively large (21,899) cohort of dialyzed patients in the United States, a Hgb concentration below 8 g/dL was associated with twofold higher odds of death in comparison to those with a Hgb concentration between 10 and 11 g/dL; several laboratory parameters related to iron status and nutrition, as well as dose of dialysis, were also associated with Hgb concentration.²²⁰ Mortality was not associated with Hgb, when Hgb values were above 11 g/dL in this cohort study, but other studies suggested that the relationship among Hgb, comorbidities, and outcomes extends into the normal range of Hgb. Other specific complications, such as proliferative retinopathy in patients with diabetes, were found to be associated with anemia.⁴³¹ Anemia has also been associated with decreased skeletal muscle mass in kidney transplant recipients, lower work productivity,⁴¹² cognitive impairment,⁴³² and increased risk of dementia.⁴³³

However, although anemia is a sensitive risk marker for many adverse outcomes, its role as a causal risk factor has not been established. Attempts to improve patient outcomes with higher Hgb targets failed apart from some positive effects on HRQOL and a reduction in transfusion requirements^{434,435} (see also later).

ERYTHROCYTOSIS OF PATIENTS WITH KIDNEY DISEASE

Although anemia is a typical complication of advanced CKD, irrespective of its cause, in a few circumstances disorders of kidney function or structure can result in abnormally high rates of RBC production—that is, erythrocytosis. In terms of kidney function, two causes of erythrocytosis in CKD have been reported. Erythrocytosis is associated with IgA nephropathy in some patients who have normal levels

of endogenous EPO for CKD,⁴³⁶ but their plasma IgA1 appears to interact with erythroblast transferrin receptors leading to an increased EPO sensitivity.⁴³⁷ Use of sodium-glucose transporter 2 inhibitors⁴³⁸ leads to higher levels of endogenous EPO that may be due to relatively higher oxygen consumption in the kidneys, induction of liver EPO production, or another unrecognized mechanism.⁴³⁹

(POLY)CYSTIC KIDNEY DISEASE

The degree of anemia in patients with autosomal dominant polycystic kidney disease (ADPKD) is usually less severe than for other causes of CKD, although patients with ADPKD on dialysis usually require treatment with ESAs. Occasionally, patients with ADPKD may become polycythemic.⁴⁴⁰ Erythrocytosis may also develop in patients on hemodialysis with acquired renal cysts and single cysts.^{441,442} Serum EPO concentrations in patients with ADPKD are, on average, approximately twofold higher than in patients with CKD from other causes,^{236,250,443} and significant arteriovenous concentration differences for EPO have been found in polycystic kidneys.⁴⁴⁴ In the cyst walls of patients with ADPKD, interstitial cells express EPO mRNA and cysts derived from proximal but not distal tubules contain increased concentrations of bioactive erythropoietin.⁴⁴⁴ Continuous activation of HIF was demonstrated in cyst walls of patients with ADPKD and in a rat model of cystic kidney disease.⁴⁴⁵ The physiologic distinction between HIF-1 α expression in tubular cells and HIF-2 α expression in peritubular cells is maintained in the cyst walls. The genetic defects underlying ADPKD do not lead to HIF activation. However, cyst expansion results in pericystic hypoxia, and hypoxic stimulation of pericystic angiogenesis is believed to play an important role in cyst progression.^{446,447} Therefore the enhanced production of EPO in cystic kidneys is probably due to local hypoxia and mediated via HIF activation. It is possible that factors other than EPO, induced through this pathway, contribute to cyst growth. Regional hypoxia also appears to stimulate cyst growth, primarily via increased fluid secretion into the cyst lumen.^{448,449}

POSTTRANSPLANTATION ERYTHROCYTOSIS

Kidney transplantation is usually followed by full correction of CKD-related anemia.^{416,450,451} Interestingly, a regular increase of EPO production is not related to the presence of the transplant but does correlate with the onset of graft function,⁴⁵² providing further evidence for the role of excretory kidney function in EPO regulation. Some 10% to 20% of patients manifest overcorrection and demonstrate erythrocytosis, usually within the first 6 months following transplantation.⁴⁵³⁻⁴⁵⁵ Graft failure is associated with anemia, and therefore polycythemia is more likely to occur in patients with normal or near normal kidney allograft function.^{236,416}

Increased plasma EPO concentrations have been reported in patients with posttransplantation erythrocytosis.⁴⁵⁶ Selective venous catheterization studies and the response to removal of the native kidneys have suggested that the native kidneys are the main source of increased EPO production.^{456,457} Although this suggestion clearly indicates that a sufficient production capacity for EPO may be preserved in diseased kidneys, it is unclear how the secretion rate is enhanced after transplantation. Improvement of the

uremic state has been speculated to play a role. Moreover, given that inflammatory cytokines can inhibit EPO production, the application of immunosuppressive agents could theoretically enhance EPO formation. Interestingly, the prevalence of posttransplantation erythrocytosis seems to be elevated in combined kidney and pancreas transplantation,⁴⁵⁸ but whether the erythrocytosis is related to enhanced EPO formation or insulin-stimulated pathways remains unclear. In some patients with posttransplantation polycythemia, circulating EPO concentrations are normal or reduced; in these cases there may be an increased sensitivity of the erythroid progenitor cells to EPO or loss of other feedback control mechanisms.

The most effective therapy of posttransplantation erythrocytosis consists of agents blocking the RAAS.^{454,459-461} No evidence supports angiotensin acting directly on EPO-producing cells, but RAAS blockade may inhibit erythropoiesis by several mechanisms (see earlier). Alternative therapeutic strategies to reduce increased RBC concentrations after transplantation include the discontinuation of diuretics, application of theophylline,⁴⁶⁰ and phlebotomy, which, however, can lead to iron deficiency.

RENAL ARTERY STENOSIS

Although renal artery stenosis reduces the oxygen supply to the kidneys, it is only rarely associated with erythrocytosis.⁴⁶²⁻⁴⁶⁵ The data on EPO production in this case are contradictory. In experimental animals, enhancement of EPO production after renal artery stenosis has been demonstrated by some, but not all, investigators.⁴⁶⁶ A study performed in rats has shown that graded reduction of renal blood flow to 10% of the control value caused a maximal threefold increase in serum EPO concentrations.⁴⁶⁷ Therefore renal EPO production appears relatively insensitive to changes in renal blood flow. Because the ratio of oxygen demand and delivery determines local oxygen tension in the area of EPO-producing cells, it is possible that the two are equally reduced after a reduction in renal blood flow, thus not resulting in sufficient hypoxia to stimulate EPO gene expression. It has been argued that the indirect coupling of oxygen demand to supply makes the kidney an ideal site for the oxygen sensing that controls RBC production.⁴⁶⁸

RENAL TUMORS

Up to 5% of patients with renal carcinomas have erythrocytosis⁴⁶⁹ and, conversely, approximately one-third of tumor-associated erythrocytosis is caused by renal cancer.⁴⁷⁰ Conflicting data have been reported concerning serum EPO concentrations in patients with renal tumors but, at least in some patients, raised values have been found.⁴⁷¹ Furthermore, overexpression of EPO mRNA has been demonstrated in renal tumors.⁴⁷² In situ hybridization has revealed accumulation of EPO mRNA in epithelial tumor cells but not in interstitial cells of the tumor stroma.⁴⁷² Most clear cell renal carcinomas, the most frequent type of renal cancer, are associated with mutations of the VHL gene that interfere with its ability to target HIF for proteasomal degradation (see earlier).⁴⁷³ Indeed, clear cell renal carcinomas contain high concentrations of HIF.⁴⁷⁴⁻⁴⁷⁷ Although stabilized HIF in kidney tumors appears to be functionally active

in inducing HIF-target genes, it is as yet unclear why overexpression of EPO is confined to about one-third of these tumors.⁴⁷⁸ Although activation of HIF appears necessary for EPO gene expression in renal cell carcinoma (kidney cancer), it is clearly not the only determinant. That erythrocytosis occurs far less frequently than overexpression of EPO in renal cancer is probably due to a variety of mechanisms causing anemia in patients with cancer, which include inhibition of the effect of EPO and reduced iron availability. Some, albeit controversial, evidence suggests that EPO has autocrine or paracrine tumor growth-promoting effects.^{479,480}

Studies are under way to treat clear cell renal carcinoma with an agent that inhibits binding between HIF-2 α and HIF- β .⁴⁸¹ As expected, as a side effect this treatment results in worsening of anemia.⁴⁸²

Clinical Relevance

When using ESAs, one should always consider the possibility of functional iron deficiency, especially in women and men with normal or borderline normal iron stores. Serum ferritin concentrations are not indicative of iron stores in the presence of inflammation, including most patients with CKD, whether receiving dialysis or not. Functional iron deficiency is less likely when serum ferritin values are >100 for ND-CKD and >200 for patients on dialysis. After instituting ESA treatment, one should monitor hematologic response and assess iron adequacy based on Hb increase.

TREATMENT OF CHRONIC KIDNEY DISEASE-RELATED ANEMIA

ERYTHROPOIESIS-STIMULATING AGENTS

Recombinant human erythropoietin was developed in the 1980s, with support from an orphan drug program. At that time, it was unclear to what extent the anemia observed in patients with kidney disease could be ameliorated by application of the hormone, as well as how many patients might benefit from this type of therapy. The initial clinical studies revealed an unexpected efficacy in patients undergoing dialysis, with high response rates and evidence that Hgb concentrations could not just be increased to some extent but could virtually be normalized.^{92,94} In subsequent years, use of rhEPO in patients on maintenance dialysis became routine in most parts of the world. The indication was subsequently extended to the much larger group of patients with ND-CKD, as well as to several other patient groups with anemia including patients with cancer who had been treated with chemotherapy.

Over the years, the efficacy of the therapy and presumed benefits led to a gradual increase in Hgb concentrations in virtually all treated patient groups. Data from the USRDS indicated a substantial increase in the use of these agents (as well as intravenous [IV] iron and blood transfusion) in older (≥ 67 years) adults with KFRT.^{483,484} Investigators originally intended to copy the endogenous molecule as closely as possible, but patent and marketing considerations, together with concepts for improving patient management, resulted in the development of a number of derivatives of the EPO molecule with altered pharmacokinetic properties and, later,

the development of different molecules that can directly or indirectly stimulate the EPO-R. The term “erythropoiesis-stimulating agents” (ESA) is commonly used to describe the heterogeneous class of drugs that increase erythropoiesis through direct stimulation of the erythropoietin receptor.

Epoetin

The term *epoetin* is usually applied to rhEPO preparations, produced by means of expression of the human EPO gene in mammalian cell lines. Production in mammalian cells rather than bacteria is required because EPO is a highly glycosylated molecule, and bacteria lack the ability to generate glycoproteins. Epoetin alfa and epoetin beta are the two agents first developed by two different companies. Both are produced in Chinese hamster ovary cells and show a high degree of similarity, with identical protein backbones of 165 amino acids and one O-linked and three N-linked glycosylation sites each, but with subtle differences in their carbohydrate composition.⁴⁸⁵

Although the amino acid sequence unequivocally determines glycosylation sites, the precise composition of the sugar side chains is also determined by the repertoire and activity of glycosylating enzymes, which may vary among cell lines and under different tissue culture conditions. Glycosylation of the EPO molecule is not required for binding or activation of its receptor¹⁰⁴; in fact, the in vitro activity of deglycosylated EPO is enhanced.⁴⁸⁶ In vivo, however, deglycosylated EPO is inactive because of rapid clearance from the circulation, indicating that the carbohydrate chains are responsible for EPO's pharmacokinetic and pharmacodynamic properties.

Early clinical trials in patients on hemodialysis used IV epoetin administered three times weekly; subsequently, the intraperitoneal (IP), subcutaneous (SC), and intradermal routes of administration were also investigated.^{487,488} After IV administration, plasma EPO concentrations decay monoexponentially, with an elimination half-life of approximately 4 to 11 hours.⁴⁸⁹ The apparent volume of distribution of EPO is about one to two times the plasma volume, and the total body clearance is lower than for other protein hormones, such as insulin, glucagon, and prolactin. The IP route was investigated as a potential means of administering EPO to patients on peritoneal dialysis, but the bioavailability of IP epoetin is disappointingly low, 3% to 8%. This application has, therefore, not been pursued.^{487,490,491}

With SC administration, peak serum concentrations of 4% to 10% of an equivalent IV dose are obtained at around 12 hours and thereafter decay slowly, so concentrations above baseline are still present at 4 days.^{488,489} The bioavailability of SC epoetin is about 20% to 25%. Nevertheless, SC application is more efficient than IV application, allowing a dose reduction of approximately 30% to maintain the same hemoglobin concentration.^{492,493} Presumably, the early peak concentrations of epoetin after IV injection are inefficient, and the more prolonged elevation of hormone levels following SC application allows a more sustained stimulation of RBC production. Thrice-weekly administration has remained the most frequent dosage for both IV and SC administration, although once-weekly,⁴⁹⁴ twice-weekly, and seven times weekly (once-daily) dosing have all been used.⁴⁹⁵ With IV epoetin, once-weekly administration is associated with much lower efficacy as compared to twice- or thrice-weekly dosing.

A number of additional epoetin preparations have been developed worldwide. Some have distinct differences in the production process—for example, epoetin delta was produced in a human cell line through increased transcription of the endogenous EPO gene,⁴⁹⁶ but this product is currently not being distributed. Other epoetins are so-called “biosimilar” generic drugs designed as copies of epoetin alfa or beta; these biosimilars are being licensed on the basis of a more limited clinical trial program after expiration of the patents for the original medications in Europe.⁴⁹⁷ Additional epoetins are available in other parts of the world, which are not necessarily produced to the same regulatory standards as the preparations marketed in the United States and Europe and may show variable product characteristics.⁴⁹⁸ Various studies in CKD patients have compared biosimilar and originator ESAs, showing no significant difference in safety or effectiveness.⁴⁹⁹⁻⁵⁰⁵

The importance of the formulation of epoetins was highlighted in 2002 with an upsurge in cases of antibody-mediated, pure red cell aplasia (PRCA) in association with the SC use of epoetin alfa marketed outside the United States after a change to an albumin-free formulation. Patients affected by this complication developed neutralizing antibodies against both rhEPO and the endogenous hormone, resulting in severe anemia and transfusion dependence.⁵⁰⁶ The cause of this serious complication remains obscure, although circumstantial evidence suggested that rubber stoppers of prefilled syringes used for the albumin-free epoetin alfa formulation may have released organic compounds that act as immunologic adjuvants.⁵⁰⁷ Factors such as breach of the cold storage chain may also have played a role. In anti-EPO antibody cases observed so far, the SC application route was usually a prerequisite. Although the unfortunate combination of adverse factors leading to a temporary increase in antibody-induced PRCA was specific for one product, a low baseline rate of PRCA persists with use of epoetin beta and darbepoetin alfa (see later).⁵⁰⁸

Darbepoetin alfa

Darbepoetin alfa is an EPO derivative with two additional bioengineered N-linked glycosylation sites, created by site-directed mutagenesis to prolong its plasma survival time (as discussed earlier).¹⁰⁶ Each of these glycosylation sites can carry an additional four sialic acid residues, such that darbepoetin alfa contains five N-linked and one O-linked glycosylation chains and has the capacity to display as many as 22 sialic acid residues, compared with a maximum of 14 sialic acid residues for original rhEPO. The additional glycosylation on darbepoetin alfa results in a molecular weight of 37.1 kDa compared with 30.4 kDa for epoetin. As intended, darbepoetin alfa has a longer half-life in vivo than rhEPO—25.3 hours versus 8.5 hours after IV administration.¹⁰⁸ The elimination half-life after SC administration is about 48 hours, which is approximately twice that previously reported for epoetin alfa or beta. A number of studies have examined once-weekly and once-every-other-week dosing.^{509,510} Darbepoetin alfa can both increase and maintain Hgb at these dosing frequencies, and its side effect profile is similar to that of epoetin alfa or beta.^{487,511} Several “conversion” studies have suggested that an appropriate conversion factor for switching patients on

epoetin alfa or beta to darbepoetin alfa is 200 U of epoetin to 1 µg of darbepoetin alfa. In contrast to epoetin alfa or beta, the dose requirements for darbepoetin alfa do not differ significantly between the IV and SC administration routes.

Methoxypolyethylene glycol epoetin beta

Alternative bioengineering techniques that further prolonged the half-life of EPO led to the development of methoxypolyethylene glycol epoetin beta (also called “continuous erythropoietin receptor activator” [CERA]), a PEGylated derivative of epoetin beta with an elimination half-life of about 130 hours when administered IV or SC.⁵¹²⁻⁵¹⁴ A methoxypolyethylene glycol polymer chain is integrated through amide bonds between the N-terminal amino group or the ε-amino group of lysine (predominantly lysine-52 or lysine-45), with a single succinimidyl butanoic acid linker. The molecular weight of CERA is twice that of epoetin (~60 kDa). Phase 3 studies have shown that, due to its longer half-life, CERA given IV once every 2 weeks was as safe and effective as epoetin given thrice weekly for correcting anemia in patients on hemodialysis.⁵¹⁵ A larger study has also shown that CERA given at 4-week dosing intervals was not inferior to epoetin given three times weekly in terms of maintaining Hgb concentrations.⁵¹⁶ No significant differences in outcomes or complications have emerged for CERA compared with other ESAs.⁵¹⁷ Conversion factors have been established and tested for switching from other ESAs to CERA.⁵¹⁸

Other erythropoiesis-stimulating agents

Several other ESAs have been developed including EPO polymers, EPO fusion proteins, and the previously mentioned EPO mimetic molecules.⁵¹⁹⁻⁵²² The ability of molecules that are structurally unrelated to EPO to dimerize the EPO-R and activate the intracellular signaling cascade was first described 30 years ago.⁵²³ Peginesatide, an EPO mimetic peptide developed for the treatment of anemia, has an amino acid sequence unrelated to that of native or rhEPO, although it shares the same biologic properties as EPO with respect to EPO-R activation.^{524,525} The potential advantages of peginesatide include greater ex vivo stability allowing storage at room temperature, prolonged pharmacodynamic action allowing once-monthly administration, and a simpler manufacturing process involving synthetic peptide chemistry. In addition, because it was structurally unrelated to EPO, peginesatide did not cross-react with anti-EPO antibodies, allowing effective treatment of anti-EPO antibody-mediated PRCA.⁵²⁶ Two phase 3 studies demonstrated that peginesatide was not inferior to conventional epoetin for correcting anemia in CKD.^{527,528} However, for reasons that remain unclear, peginesatide increased the risk of a combined cardiovascular endpoint in patients with ND-CKD.⁵²⁷ Accordingly, it was only approved in the United States and for use in patients on dialysis only. Slightly more than 6 months after its introduction, the drug was recalled due to postmarketing reports of serious hypersensitivity reactions—including fatal reactions in approximately 0.02% of patients—that occurred within 30 minutes after the first IV dose and had not been reported during clinical trials.⁵²⁹ Currently, no EPO mimetic is available for rescue therapy in patients with anti-EPO, antibody-mediated PRCA.

Initiation and maintenance of erythropoiesis-stimulating agent therapy

Following commencement of therapy with ESAs or HIF-PHIs (see below), a significant increase in the reticulocyte count is evident at 2 weeks and an increase in Hgb concentration is seen by 2 to 3 weeks.⁵³⁰ The increases are dose-dependent, and most physicians aim for an Hgb increment of not more than 1 g/dL/month to minimize the risk of adverse effects. In most patients, ESA therapy is initiated at the outset of dialysis therapy and, according to the USRDS report, peak doses of ESA are being administered at month 2 after the initiation of dialysis. Only 0.3% of U.S. patients with ND-CKD are treated with ESAs; these patients are usually older, with more advanced disease and significant comorbidities, resulting in high rates of death and cardiovascular complications.⁵³¹

The increase in Hgb concentration following ESA therapy results from an increase in RBC count. No significant changes in leukocyte or platelet counts are usually seen, although a moderate increase in the platelet count has been documented in some studies. Serum ferritin concentration and/or the transferrin saturation are decreased after the initiation of ESA therapy, unless iron stores are being replenished in parallel, because large quantities of iron are used up in the manufacture of new RBCs (see later).

Radioisotope blood volume studies have confirmed that there is an increase in RBC mass after treatment with ESAs, associated with a compensatory reduction in plasma volume so that the whole blood volume remains unchanged. Early ferrokinetic studies have indicated that epoetin therapy induces a twofold increase in marrow erythropoietic activity, as evidenced by a doubling of marrow and RBC iron turnover.^{249,532}

HYPOXIA-INDUCIBLE FACTOR STABILIZERS

The HIF stabilizers are competitive inhibitors of HIF prolyl hydroxylases and asparagyl hydroxylase, enzymes involved in the degradation of HIF and suppression of its transcriptional activity, respectively (as discussed earlier).^{52,533,534} The HIF stabilizers, also termed HIF prolyl hydroxylase inhibitors (HIF-PHIs), increase endogenous EPO production.^{535,536} These drugs are orally active. Given their entirely different mechanism of action, HIF-PHIs are not subsumed under the term ESA.⁵³⁷⁻⁵³⁹

An early, phase 1/2, single-dose study comparing endogenous EPO production in small groups of patients on dialysis with native kidneys or after bilateral nephrectomy provided evidence that EPO production can be stimulated in extrarenal sites (presumably the liver) and the diseased, nonfunctioning, fibrotic kidneys.²⁴⁷ These data provided proof for the concept that a disturbance of the renal oxygen-sensing mechanism, rather than a loss of EPO-producing cells, is the main cause of CKD-related anemia.

Several HIF-PHIs have been developed, tested in phase 3 programs, and approved in many countries for treatment of CKD-related anemia.^{538,539} The motivation driving their development included the ease of oral administration, easier production and increased stability of these small molecules as compared with the ESA biologicals, as well as potentially synergistic effects related to increased iron availability.

Given the previously mentioned safety issues related to targeting higher Hgb concentrations with ESAs, lower plasma EPO concentrations achieved by subtle, intermittent activation of endogenous EPO production should be theoretically safer than the high, supraphysiologic concentrations following intravenous bolus injections of ESAs. However, phase 3 programs do not support this assumption of improved safety. While studies comparing different HIF-PHIs with ESAs have overall shown similar efficacy, HIF-PHIs were associated with increased rates of major cardiovascular events and other adverse outcomes in several studies.^{6,538,539} Specifically, safety was lower in patients with ND-CKD, the population in which the advantage of oral administration would be most relevant.^{538,540-543} Therefore two medications with large development programs, daprodustat and vadadustat, have only been approved in the United States and Europe for patients on dialysis for at least 3 to 4 months. Another medication, roxadustat, was approved in Europe and many non-European countries, but not in the United States due to safety concerns.^{6,539} No head-to-head comparisons of different HIF-PHI are available, but study data indicate that different agents have different profiles and risk/benefit ratios need to be assessed separately. Clinical trials suggest that HIF-PHIs increase iron availability, as expected,⁵³⁰ but effects on iron metabolism and/or need for iron therapy were not endpoints in clinical trials. Whether HIF-PHIs are more effective than ESAs in people with ESA hyporesponsiveness remains unclear.⁵³⁹

IRON MANAGEMENT

Iron is the fourth most common element—after oxygen, silicon, and aluminum—in the earth's crust and the most abundant transitional metal in the human body. Although it plays an essential role in multiple biologic processes, such as transport of oxygen, transfer of electrons, DNA synthesis, and heme-based enzymatic reactions, iron is also highly susceptible to transition from the ferrous state (Fe^{2+}) to the ferric (Fe^{3+}) state and to generation of reactive oxygen species (ROS) via the Haber-Weiss-Fenton reaction, thus requiring multiple systems to prevent or control this potentially harmful transition.²⁰⁴

The metabolism of iron is geared toward conservation and recycling, with the duodenal absorption of iron in adults being tightly regulated to compensate for the small daily losses and, thereby, keeping the total iron pool constant, in the range of 35 to 45 mg per kilogram of body weight. Approximately two-thirds of body iron is contained in the RBC pool as Hgb, with much of the remaining fraction of the total pool in the heme of muscle myoglobin or stored in reticuloendothelial system (RES) macrophages and the liver.

Multiple factors decrease total body iron or restrict iron availability in patients with CKD, including reduced intake and absorption, chronic losses due to occult and overt bleeding, and reduced bioavailability of iron due to the chronic inflammatory state and higher hepcidin concentrations (see previously).⁵⁴⁵ Iron losses in patients with CKD can be up to 5 to 6 mg iron daily (1 mg iron daily in normal subjects) and cannot be adequately compensated with oral iron supplements because duodenal absorption and release from macrophages is limited by the chronic inflammatory state and associated elevated hepcidin. In

addition, in patients with CKD treated with ESAs, insufficient amounts of iron are released from the body stores to meet the greater demand of ESA-driven erythropoiesis.⁵⁴⁶ Similar evidence has been provided for persons with normal kidney function when erythropoiesis is increased by an intensive blood donation schedule and EPO, or EPO alone, despite concomitant oral iron supplementation.⁵⁴⁷⁻⁵⁴⁹

MARKERS OF IRON STATUS

When hematologic signs of iron-deficient erythropoiesis—reduced Hgb concentration with abnormally low MCV and MCH values, inadequate reticulocyte response, and low reticulocyte Hgb content—are associated with biochemical markers of low iron stores (abnormally low serum ferritin), the diagnosis of absolute iron deficiency is straightforward. However, straightforward determination of iron deficiency is the exception rather than the rule for patients with CKD, in whom iron may be present in storage form but not readily available for erythropoiesis, and serum ferritin concentrations are increased owing to the concomitant inflammatory state (see earlier). Thus the diagnosis of iron deficiency in CKD must rely on a variety of markers, biochemical and hematologic and, in the most challenging cases, on the erythroid response to IV iron (see later for limitations of bone marrow biopsy). These markers are determined in individual patients over time; markers with high biologic and analytic variability, such as transferrin saturation and ferritin, are less suitable to assess iron status than markers with low variability such as Hgb, Hct, and reticulocyte Hgb content.^{550,551}

Serum ferritin

Serum ferritin values higher than 200 µg/L are recommended for patients on dialysis, and values of 100 µg/L should be considered the lower limit of normal for patients with ND-CKD.²⁰ The sensitivity for ruling out iron deficiency was reported to be 90% for a ferritin cutoff of 300 µg/L and 100% for 500 µg/L.^{552,553} Concerns about iron toxicity and overload have resulted in several guidelines setting an upper limit for ferritin of 500 to 700 µg/L,^{10,45} above which IV iron is not recommended. Additional factors that elevate serum ferritin are hyperthyroidism, liver disease (associated with hepatitis C virus [HCV] or other conditions), alcohol consumption, and oral contraceptives, whereas vitamin C deficiency and hypothyroidism decrease ferritin concentrations.⁵⁵⁴ More than 50% of U.S. patients on hemodialysis have serum ferritin values higher than 800 ng/mL, and more than 20% have values >1200 ng/mL.

Serum iron, transferrin, and transferrin saturation

The biochemical markers serum iron, transferrin (measured as TIBC), and transferrin saturation (TSAT; serum iron/serum TIBC) are routinely used in the diagnosis of iron deficiency but have some important limitations. Serum iron concentrations and TSAT values are sensitive to diurnal variations and to dietary intake, with serum iron concentrations being higher early in the day. These are increased by greater iron intake with food or dietary supplements and decreased in the presence of infection and inflammation, but with more severe inflammation, the TSAT increases due to declining

TIBC.⁵⁵⁵ Some biochemical methods used to measure iron are sensitive to hemolysis and will produce falsely elevated iron values,⁵⁵⁶ whereas other serum iron assays have been shown to perform poorly in patients on dialysis.⁵⁵⁷ Serum transferrin can be elevated by the use of oral contraceptives and reduced with inflammation or infection. Several studies have shown that these traditional biochemical iron parameters perform poorly in CKD and are inferior to some of the newer hematologic parameters (described later).⁵⁵⁸⁻⁵⁶¹ However, a lower serum iron concentration has been shown to be an independent predictor of mortality and hospitalization in people on dialysis,⁵⁶² and a higher TSAT value has been associated with lower mortality.⁵⁶³ A TSAT of 20% is generally considered a threshold value below which iron therapy is indicated.^{5,6,539} In one study using data from NHANES, more than 50% of the noninstitutionalized adult U.S. population was found to have values below the CKD threshold for ferritin (100 ng/mL) and TSAT (20%).⁵⁶⁴ Overall, women were more likely to have laboratory-based evidence of iron deficiency. Men with CKD had a higher prevalence of iron deficiency than men without CKD, whereas the prevalences in women with and without CKD were similar. Serum transferrin and total iron-binding capacity (TIBC) are dual markers of iron status and of nutritional status and protein balance: A lower baseline TIBC value or its decrease over time in dialyzed patients is associated with higher mortality, protein-energy wasting, and worsening inflammation.⁵⁶⁵

Soluble transferrin receptor

Soluble transferrin receptor (sTfR) concentration in the serum is a marker of iron status as transferrin receptors are shed from the membrane of maturing erythroblasts and reticulocytes, either in soluble form or as vesicles.⁵⁶⁶⁻⁵⁶⁸ The concentration of sTfR is abnormally elevated in iron deficiency as iron-deficient erythroblasts overexpress TfRs in the plasma membrane via IRP stabilization of TfR mRNAs.²⁰⁴ sTfR concentration is not affected by inflammation, but as the total number of marrow erythroblasts increases, such as in hemolytic anemias or treated deficiencies of iron, folate, or EPO, the pool of maturing erythroblasts increases, making interpretation more complex.⁵⁶⁹⁻⁵⁷³

A study in patients on dialysis with anemia has shown that a sTfR concentration lower than 6 mg/L (which rules out iron deficiency; normal value, 3.8–8.5) was associated with responsiveness to the initiation of rhEPO therapy.⁵⁷⁴ However, because increased erythropoiesis by itself raises the sTfR concentration, sTfR measurement could not reliably detect functional iron deficiency in patients on maintenance ESA therapy. Other studies have failed to show a predictive value for sTfR in CKD anemia management.^{575,576} A decline in sTfR concentration may reflect increases in iron availability when IV ascorbic acid is used to mobilize iron stores.⁵⁷⁷ Race and ethnicity, smoking, alcohol consumption, and body mass index have been shown to be associated with sTfR values.^{578,579} The sTfR/ferritin ratio provides a more accurate assessment of iron status and the need for iron supplementation,^{567,580,581} Clinical laboratory values are often converted to the sTfR/ferritin index (sTfR[mg/L]/log ferritin[µg/L]), but clinical use has been limited including in patients on dialysis.⁵⁸²⁻⁵⁸⁴

Clinical Relevance

Although oral iron may be effective in ND-CKD patients, the response is generally suboptimal, as many patients have gastrointestinal side effects. Side effects for the various IV iron preparations available in the United States and European markets are usually rare and limited to mild to moderate hypersensitivity reactions. In approximately 0.05% of IV iron administrations, arthralgias, myalgias, or flushing appear at the beginning of the infusion in the absence of associated hypotension, tachypnea, tachycardia, wheezing, stridor, or periorbital edema. No intervention is necessary except for a temporary halt in the IV infusion. After symptoms abate, IV infusion can be restarted.

Percentage of hypochromic red blood cells

A distinguishing characteristic of iron-deficient erythropoiesis is the HRI-mediated production of hypochromic microcytic reticulocytes that mature into hypochromic microcytic erythrocytes (see earlier). Iron-deficient erythropoiesis results in an increase in the percentage of hypochromic erythrocytes (%HYPO), defined as the percentage of erythrocytes with a mean corpuscular hemoglobin concentration (MCHC) lower than 28 g/dL, for hematology analyzers produced by Siemens Medical Solutions.^{585,586} Similar parameters (low hemoglobin density, [LHD%], and DF-Hypo XE, respectively) are available when using Beckman-Coulter and Sysmex instruments.^{587,588} A study by Macdougall and colleagues⁵⁴⁶ has first shown that functional iron deficiency induced by epoetin treatment and the response to IV iron could be detected by changes in %HYPO. Several studies have confirmed that an increased %HYPO is a sensitive and early indicator of iron deficiency.^{559,589-593} A European study has found %HYPO to be the only independent predictor of mortality among various iron status parameters, with a twofold higher mortality risk for values higher than 10% than for values lower than 5%.⁵⁹⁴ A clinical study has suggested a %HYPO target of <10% by prospectively raising the delivered dose of IV iron to 228 patients to achieve a %HYPO lower than 2.5% and a serum ferritin concentration of 200 to 500 ng/mL.⁵⁹³ In this study, the median %HYPO value decreased from 8% to 4%, the median serum ferritin concentration increased from 188 to 480 ng/mL, and the median rhEPO dose decreased from 136 to 72 IU/kg per week, showing that a strategy aimed at achieving %HYPO values much lower than 10% could be cost-effective. However, it also resulted in serum ferritin values in some patients much higher than those recommended by guidelines.

Contrary to the European studies, North American studies have failed to show value for %HYPO in assessing iron availability in dialysis patients.^{595,596} The reasons for this discrepancy are not clear. It is worth noting that %HYPO progressively increases with storage of the blood sample, owing to the concomitant increase in MCV and reduction in MCHC, and is therefore best measured within 4 hours. In addition, %HYPO increases with reticulocytosis because reticulocytes have a lower MCHC than mature RBCs.⁵⁹⁷ A new automated determination of low hemoglobin (LDH%)

based on the MCHC can distinguish iron deficiency in the setting of ACD and the anemia of CKD.³⁵⁷

Reticulocyte hemoglobin content

After being released from the marrow, reticulocytes spend 18 to 36 hours in the circulation before becoming mature erythrocytes. Studies of the cellular characteristics of reticulocytes thus provide a real-time assessment of the functional state of the bone marrow. Automated analyzers can determine not only the absolute number of reticulocytes with great precision, but also their size and Hgb concentration and content.^{598,599} The reticulocyte Hgb content⁶⁰⁰ (CHr or RetHb; in pg/cell) has been extensively studied, especially in patients treated with rhEPO.⁶⁰¹ A reduction in CHr is the most sensitive indicator of functional iron deficiency. Healthy subjects with normal iron stores who were treated with rhEPO produced a substantial fraction of hypochromic, low Hgb content reticulocytes when their baseline serum ferritin values were below 100 µg/L.⁵⁴⁸ When IV iron was used in conjunction with rhEPO in normal subjects, the production of hypochromic reticulocytes was absent.⁶⁰² Several small studies have described the value of CHr in identifying iron deficiency in patients on dialysis, mostly based on the subsequent response to IV iron.^{561,595,596,603} A sensitivity of 100% and specificity of about 70% to 80% were reported in one study,⁵⁹⁶ although other studies have reported lower values.^{561,603} These initial studies led to additional large clinical trials that tested the values of CHr in managing the dosing of IV iron and rhEPO in dialysis patients. Fishbane and colleagues⁵⁶⁰ randomly assigned 157 patients to two different IV iron management strategies, one based on CHr, in which IV iron was started if CHr fell below 29 pg/cell, and one in which IV iron was started if the serum ferritin fell below 100 ng/mL or the TSAT values decreased below 20%. A significant reduction in exposure to IV iron was obtained in the CHr-based management, with no differences in weekly rhEPO dosing between the two groups.⁵⁶⁰ Tessitore and coworkers⁵⁵⁹ have compared the diagnostic precision of a variety of hematologic and biochemical markers to identify patients who exhibit an increase in Hgb values in response to IV iron. A combination of %HYPO higher than 6% and CHr less than 29 pg/cell has shown the best diagnostic efficiency for iron deficiency (80%) based on the Hgb response to IV iron. Other studies have provided additional confirmation of the diagnostic value of CHr,^{604,605} although one has questioned its superiority to TSAT,⁶⁰⁶ and only one study has shown that the use of IV iron in patients with low CHr resulted in decreased weekly usage of rhEPO.⁶⁰⁷ A normal reticulocyte Hgb content may be used to optimize the use of IV iron.⁶⁰⁸

Several studies have also validated reticulocyte Hgb measurements (RET-Hb and Ret-Hb) generated by analyzers produced by Sysmex.^{550,588,609-612} Guidelines on the diagnosis and management of iron deficiency have included reticulocyte Hgb among those recommended.^{551,613} A limitation of reticulocyte Hgb is that it cannot be used to assess iron availability in the presence of thalassemia traits (α or β) or megaloblastic erythropoiesis.

Bone marrow iron

Although iron staining of a bone marrow biopsy is regarded as the gold standard method of assessing iron stores, widely

divergent estimates of the prevalence of iron deficiency have been generated by this invasive, potentially painful procedure.^{553,614-616} A study in 100 patients with ND-CKD has shown that evaluation of iron stores by iron staining of a bone marrow sternal aspirate was not better than TSAT or ferritin in correctly identifying responders to IV iron therapy.^{617,618} Some patients with CKD have been found to have no stainable iron evident on sternal bone marrow biopsy, despite the presence of normal to elevated serum ferritin concentrations.^{616,619} Bone marrow studies also have no value in identifying patients at risk for the development of functional iron deficiency with EPO therapy.

Liver magnetic resonance imaging

Hepatic magnetic resonance imaging (MRI) provides a noninvasive tool for monitoring patients with iron overload disorders. However, the number of studies applying this technology in CKD is still limited.⁶²⁰⁻⁶²³ In the largest of these studies, conducted in 119 patients undergoing hemodialysis and receiving IV iron in a single center, according to current guidelines, 84% had evidence of hepatic iron deposition and 30% had hepatic MRI findings consistent with severe iron overload.⁶²² These data raise concerns that the use of IV iron and current thresholds for laboratory parameters may be too liberal, especially considering the increased use of IV iron in the United States since 2011. On the other hand, contrary to cardiac iron MRI assessments,⁶²⁴ it has not yet been demonstrated that the observed increases in hepatic iron are of any functional significance and/or are associated with clinically relevant adverse outcomes.^{273,625}

Iron balance considerations

As noted, 1 mL of blood normally contains 0.5 mg of iron and proportionately less when the Hgb concentration is reduced. An estimated annual blood loss of 2 L in a dialysis patient with moderate anemia (20% reduction in Hgb) therefore corresponds roughly to 0.8 g of iron loss. Irrespective of all parameters of iron metabolism, IV iron supplementation in excess of this amount results in a positive iron balance unless blood loss (and thereby iron loss) is more pronounced than anticipated. When patients were categorized according to their level of hepatic iron deposition, the average monthly iron doses were 150 and 283 mg in those with signs of mild and moderate iron overload, respectively, compared with 100 mg in those without.⁶²² The functional consequences of a positive iron balance, and in particular whether it carries risk, however, are still unclear.²⁷³

ORAL IRON THERAPY

While intravenous rather than oral iron therapy is recommended in patients with anemia and CKD receiving hemodialysis, patients with anemia and CKD not receiving dialysis or treated with peritoneal dialysis have the choice between intravenous and oral iron therapy. Oral iron therapy is likely to lead to a slower increase in Hgb concentrations and iron parameters and is associated with more frequent, but less severe side effects, in particular constipation and other gastrointestinal symptoms.⁶ Oral iron is also less expensive than intravenous iron, easier to use, and does not require frequent IV cannulation, which preserves veins for potential arteriovenous fistulas—lifelines for hemodialysis. Several oral iron formulations are available including

ferrous sulfate and ferrous fumarate. Ferrous gluconate, liposomal iron, and heme iron polypeptide may have improved efficacy and tolerability.⁶ Ferric citrate is an oral iron formulation and phosphate binder approved to treat iron deficiency anemia in patients with CKD not receiving dialysis.⁶²⁶⁻⁶³⁰ It also improves iron parameters and reduces ESA and IV iron exposure in patients on hemodialysis.⁶³¹ Control of serum phosphorus with another ferric citrate-based phosphate binder has been shown to be associated with lower ESA requirements and lower use of IV iron.⁶³² Sucroferric oxyhydroxide, another iron-containing U.S. Food and Drug Administration (FDA)-approved phosphate binder, has shown no significant effects on IV iron metabolism and anemia,⁶³³ indicating less efficient iron absorption than with ferric citrate.

INTRAVENOUS IRON THERAPY

Oral iron therapy is recognized as being insufficient to support the functional needs of ESA-stimulated erythropoiesis in patients with kidney failure treated by dialysis. A systematic review and meta-analysis have shown that the Hgb response is much stronger with IV iron than with oral iron, and this effect is more substantial in patients on dialysis than in patients with ND-CKD.^{634,635} A systematic Cochrane analysis has identified significant associations of IV iron therapy with increased Hgb, ferritin, and TSAT values, as well as reduced ESA requirements, with no differences in mortality.⁶³⁶ These findings were confirmed in a meta-analysis that showed a 23% dose reduction in ESA with the use of IV iron.⁶³⁷ The cost-effectiveness of IV iron therapy has also been demonstrated under the assumption that a higher mortality risk is associated with Hgb values <9.0 g/dL.⁶³⁸

Nevertheless, oral iron may be effective with patients with ND-CKD (see earlier).⁶³⁷ The FIND CKD study (Ferinject assessment in patients with iron deficiency anemia and ND-CKD) compared the efficacy and safety of oral iron with the IV administration of ferric carboxymaltose that targeted two different serum ferritin ranges, 100 to 200 and 400 to 600 µg/L.⁶³⁹ The IV therapy targeting the higher ferritin range showed greater efficacy in increasing Hgb but no difference in decreased Hgb concentrations or the need to switch to another anemia therapy between the IV arm targeting the lower ferritin range and the oral iron therapy arm. However, the response to oral iron was suboptimal in many patients, with only 21.6% showing an Hgb increase of at least 1 g/dL after 4 weeks of oral iron therapy and more than half of the patients never achieving this level of response throughout the trial.⁶⁴⁰ A systematic review and meta-analysis of trials comparing oral and IV iron therapy in CKD has shown significant benefit for IV iron use in CKD stages 3 to 5.⁶⁴¹

Several IV iron preparations are available for clinical use in the U.S. and European markets^{6,642} (Table 53.1), with most of them containing iron associated with a carbohydrate shell. The strength or lability of this association is crucial for dosing, with the most stable preparations, such as iron dextran, being suitable for large dose replacements, and the more labile preparations, such as iron gluconate, requiring multiple dosing, with a single-dose maximum of approximately 125 mg.⁶ Intravenous iron infusion may lead to some immediate binding of the infused iron to transferrin, resulting in its complete saturation and the

Table 53.1 Intravenous Iron Preparations

Generic Name	Iron Dextran	Iron Sucrose	Na Ferric Gluconate	Ferric Carboxymaltose	Ferumoxytol	Ferric Derisomaltose
U.S. trade name (marketed by)	INFed (Actavis)	Venofer (American Regent, Luitpold Pharmaceuticals)	Ferrlecit (Sanofi-Aventis); Nulecit (Actavis)	Injectafer (Luitpold, American Regent)	Feraheme (AMAG Pharmaceuticals)	Monofer (Pharmacosmos A/S)
Trade name, Europe	Cosmofer, Uniferon, Ferrisat	Venofer, Idafer, FerroLogic, Ferion, Venotrix, Fermed, Netro-Fer	Ferrlecit/Ferlixit	Injectafer, Ferinject	Rienso	Monofer, Monover, Monoferro, Diafer
Carbohydrate	Dextran polysaccharide (LMW)	Sucrose	Gluconate	Carboxymaltoside	Polyglucose sorbitol carboxymethylether	Isomaltoside
Molecular weight	165 kDa	34-60 kDa	289-444 kDa	150 kDa	750 kDa	150 kDa
Iron, mg/mL	50	20	12.5	50	30	50 or 100
Hemodialysis, mg/session	100	100	125	—	510	100-200 (UK)
Peritoneal dialysis	100	1 × 300 mg; 1 × 300 mg after 14 days; 1 × 400 mg after 14 days	—	—	510	—
CKD, nondialysis	100 mg	200 or 500 mg	—	750 mg	510 mg	—
TDI possible	Yes	No	No	Yes	No	Yes
Maximum approved dose	100 mg	400 mg	125 mg	750 mg for body weight >50 kg	510 mg	Up to 20 mg/kg (UK)
Max safe dose	TDI over 1-4 hrs	300 mg over 2 hrs	250 mg over 1 hr	750 mg over 15 min	510 mg in >15 min	20 mg/kg over 15 min
Premedication	No	No	No	No	No	No
Test dose required	Yes	No	No	No	No	No
Black box warning (FDA) ^a	Yes	No	No	No	Yes	Na
Adverse reactions	—	—	—	Hypophosphatemia	Alteration of MRI scan	—
Preservative	None	None	Benzyl alcohol	None	None	None

^aFDA warnings—iron dextran and ferumoxytol, fatal allergic reactions.

CKD, Chronic kidney disease; FDA, U.S. Food and Drug Administration; LMW, low molecular weight; MRI, magnetic resonance imaging; NA, not applicable; TDI, tolerable daily intake.

"All intravenous iron medicines have a small risk of causing allergic reactions, which can be life-threatening if not treated promptly." European Medicines Agency. New recommendations to manage risk of allergic reactions with intravenous iron-containing medicines.

http://www.ema.europa.eu/ema/index.jsp?curl=pages/news_and_events/news/2013/06/news_detail_001833.jsp&mid=WC0b01ac058004d5c1.

subsequent generation of free iron, which has vasoactive effects and can produce hypotensive and/or anaphylactoid reactions. This risk involves mainly semilabile iron-sugar complexes such as iron sucrose and iron gluconate, and not more stable complexes, such as ferric carboxymaltose, ferumoxytol, and iron dextran.

Lower-molecular-weight iron dextran

This preparation has a significantly better tolerability and fewer side effects than the higher molecular weight product,

which has been removed from the U.S. and European markets.⁶⁴³⁻⁶⁴⁹

Iron sucrose

Iron sucrose is used worldwide to treat CKD-related anemia and is the most widely used parenteral iron preparation in the United States.⁶⁵⁰ The maximum recommended single dose is 200 mg. Allergic reactions have been reported in fewer than 1/100,000 infusions. IV injection into rats of three different commercial preparations of iron sucrose

have resulted in different degrees of inflammation and oxidative stress, suggesting that the stability of the iron complex may differ from one iron sucrose preparation to another.⁶⁵¹

Ferric gluconate

Ferric gluconate is another commonly prescribed IV iron preparation in the United States and is frequently used worldwide in patients on hemodialysis.⁶⁵² The maximum recommended single dose is 125 mg.

Ferric carboxymaltose

A significant advantage of this preparation is the possibility of infusing up to 750 to 1000 mg of iron in a short time (15 minutes), with minimal side effects.⁶⁵³⁻⁶⁵⁵ Transient hypophosphatemia has been reported in patients without CKD and in those with ND-CKD treated with ferric carboxymaltose, possibly mediated by a decreased tubular reabsorption of phosphate.^{656,657} FGF23 plays an important role in the hypophosphatemia observed with ferric carboxymaltose administration. In iron deficiency, increased FGF23 production is associated with increased cleavage within osteocytes, with unchanged FGF23 plasma values and elevated values of FGF23 c-terminal fragments. Administration of ferric carboxymaltose is associated with a large increase in plasma FGF23 values, possibly due to reduced cleavage. Low serum phosphate, increased phosphate fractional excretion, and decreased serum 1,25 vitamin D values are observed for 1 to 2 weeks following IV administration of 750 mg of iron carboxymaltose at week 0 and 1.⁶⁵⁸ In ND-CKD patients with serum ferritin concentrations between 400 and 600 ng/mL (FIND-CKD trial), ferric carboxymaltose was better than oral iron in delaying and/or reducing ESA requirements.⁶⁵⁹ There was no evidence of IV iron-related kidney toxicity or accelerated decline in kidney function in the FIND-CKD trial.⁶⁵⁹

Ferumoxytol

Ferumoxytol is an iron oxide nanoparticle with a polyglucose sorbitol carboxymethylether coating designed to minimize immunologic sensitivity and release of free iron, allowing a rapid injection of a large dose of 510 mg within a minimum infusion time of 15 minutes, which can be repeated after 3 to 8 days.⁶⁶⁰⁻⁶⁶³ Efficacy and adverse events in patients with CKD were found to be similar to those of iron sucrose.⁶⁶⁴ Ferumoxytol is the only IV iron preparation possessing super magnetic properties, similar to MRI contrast agents, which may alter MRIs for up to 3 months owing to its uptake into the reticuloendothelial system.⁶⁶¹ The development of nephrogenic systemic fibrosis following gadolinium exposure (in earlier generation gadolinium-based contrast agents) prompted the use of ferumoxytol as an alternative MRI contrast agent in patients with CKD stage 4 or 5 and in dialysis-dependent CKD.^{665,666}

On the basis of studies used for FDA registration, 0.2% of treated subjects experienced anaphylaxis or anaphylactoid reactions, 3.7% had hypersensitivity-type reactions (e.g., pruritus, rash, urticaria, wheezing), 1.9% of patients had hypotension, and three patients experienced serious hypotensive reactions.⁶⁶¹ However, subsequent studies of larger datasets have not shown an increased incidence of adverse reactions for ferumoxytol compared with other

IV iron preparations in patients with CKD.⁶⁶⁷ There were no significant differences between ferumoxytol and ferric carboxymaltose regarding hypersensitivity reactions or hypotension post-IV infusion, whereas the incidence of hypophosphatemia was greatly increased for ferric carboxymaltose (38.7% vs. 0.4%).⁶⁶⁸ The Ferumoxytol for Anemia of CKD Trial (FACT) showed that ferumoxytol had a similar efficacy and safety profile to iron sucrose in the treatment of iron deficiency anemia in patients with CKD undergoing hemodialysis.⁶⁶⁹

Ferric isomaltoside

Ferric isomaltoside is based on a nonbranched carbohydrate, which does not form the typical spheroidal iron carbohydrate nanoparticle like other IV iron preparations and seems to be associated with lower immunogenic potential. It can be administered in a maximum single dose up to 20 mg/kg and has been shown to exhibit equivalent efficacy to iron sucrose in patients receiving maintenance dialysis.⁶⁷⁰

Parenteral iron administration has increased substantially in the United States, most likely because of a shift in reimbursement practices toward a bundled capitated model. Data from the Dialysis Outcomes and Practice Patterns Study (DOPPS) showed IV iron use increasing from 55% to 68% of patients on hemodialysis between 2010 and 2012; capitated payments for dialysis, including medications to correct anemia, were instituted in 2011.⁶⁷¹ Similar trends have been reported for European countries, Japan, Australia, and New Zealand.⁶⁷² A ferritin threshold of 700 ng/mL is now commonly used for prompt withholding of IV iron therapy in patients on maintenance dialysis.

SIDE EFFECTS OF INTRAVENOUS IRON

Iron sucrose, lower-molecular-weight iron dextran, and ferric carboxymaltose have excellent track records for safety and tolerability. Hypersensitivity reactions (e.g., erythematous rash, urticaria) are rare, and their intensity is usually mild or moderate. Lack of recurrence after rechallenge indicates that most of these events are not due to immunologic reactions. Severe life-threatening allergic reactions have been a major problem with higher-molecular-weight iron dextran, prompting its removal from the European and U.S. markets. The use of a test dose is no longer recommended because a negative test dose does not rule out subsequent reactions.⁶ A retrospective study by Chertow and colleagues of more than 50 million doses of IV iron found a higher risk of reactions with higher-molecular-weight iron dextran, while rates of serious events associated with lower-molecular-weight iron dextran and other forms of IV iron were less (~1/200,000).^{644,645} A study from the FDA using data obtained from the administration's Adverse Event Reporting System (AERS) and other U.S. databases was unable to provide firm data on the relative safety of the four IV preparations marketed in the United States owing to incomplete brand information on these reports, but it did confirm that allergic reactions have been reported for all brands.^{673,674} Chertow and colleagues⁶⁴⁵ estimated absolute rates of life-threatening reaction per million doses of 0.6 for iron sucrose, 0.9 for sodium ferric gluconate complex, 3.3 for lower-molecular-weight iron dextran, and 11.3 for higher-molecular-weight iron dextran. However, a later systematic review has highlighted the lack of properly conducted and

powered studies comparing adverse event rates between lower-molecular-weight iron dextran and iron sucrose.⁶⁷⁵ The amount of labile iron, which differs among the various IV iron preparations, is likely an important determinant of possible oxidative and nitrosative stress.⁶⁷⁶

The REVOKE trial aimed to assess the effect of IV iron on the progression of CKD by comparing IV iron sucrose, 200 mg once every 2 weeks for 8 weeks, with oral iron sulfate. Although there was no difference in the slope of GFR, the authors reported a higher number of serious adverse events in the IV iron arm and concluded that the risk for infections and cardiovascular events was increased.⁶⁷⁷ This unexpected result was not confirmed in FIND-CKD, and the difference between the outcomes of both studies remains unexplained.^{678,679}

The REPAIR-IDA trial has demonstrated that a regimen of two doses of 750 mg of ferric carboxymaltose in 1 week was not inferior to up to five 100-mg infusions of iron sucrose in 14 days for anemic subjects with NK-CKD.⁶⁵⁷ These larger doses of IV iron, rather than more frequent but lower maintenance doses, do not appreciably affect cardiovascular morbidity and mortality in hemodialyzed patients.⁶⁸⁰

Infection risk and intravenous iron therapy

In vitro data seem to support the notion that iron can promote bacterial growth and, at the same time, impair leukocyte function.⁶⁸¹⁻⁶⁸⁴ IV injection of iron sucrose in dialyzed patients has been associated with the dose-dependent appearance of markers of oxidant damage in lymphocytes and a decrease in plasma ascorbate and alpha-tocopherol in some studies⁶⁸⁵ but not in others.⁶⁸⁶⁻⁶⁸⁸ In addition, studies have not accounted for the fact that the capability of leukocytes to cope with oxidant damage is markedly affected by polymorphisms in glutathione S-transferase M1 (GST M1).⁶⁸⁹ Although there is indirect and inconclusive evidence for an association between iron stores and bacteremia,⁶⁸¹ most studies have failed to show an association of IV iron therapy with the risk of infection in dialyzed patients.⁶⁸²⁻⁶⁸⁴ Many studies attempting to link iron status and risk of bacterial infection have used serum ferritin, an unreliable marker of iron status in CKD, as discussed previously.^{681,690-694} One study has shown that in patients receiving more than 10 vials of 100 mg iron dextran over 6 months, there was an increased risk of death and hospitalization.⁶⁹⁵ One uncontrolled retrospective study has reported a higher incidence of bacteremia with iron sucrose than with ferric gluconate.⁶⁹⁶ Other studies have failed to show a significant effect of IV iron dosing or iron status (using serum ferritin) on bacteremia, mortality, infection, or hospitalization.^{684,697,698} On the other hand, a later observational study using a large database has found that bolus administration of higher doses of IV iron is associated with higher risks for infection-related hospitalizations and death, particularly in patients undergoing dialysis with catheters, rather than arteriovenous fistulas or grafts.⁶⁹⁹ Higher doses of IV iron have not been shown to be associated with higher mortality risk or infections.⁷⁰⁰ Despite the lack of proof of significant effects on the rates of infections, cardiac events, and mortality, long-term toxicities and, in particular, the possible consequences of oxidant damage due to free radical generation are still a concern.^{610,701}

Importantly, a trial in the United Kingdom (PIVOTAL) has compared proactive high-dose IV iron sucrose

therapy with reactive low-dose therapy consisting of monthly injections of 400 mg only if serum ferritin was <200 ng/mL and TSAT <20% in 2141 patients on dialysis and assessed the impact of both regimens on mortality and cardiovascular events.⁷⁰² The high-dose IV sucrose regimen yielded a significant decrease in monthly ESA dose and was superior with regard to a combined endpoint of mortality and cardiovascular events.⁷⁰² The high-dose and low-dose regimens had similar infection rates.^{702,703} On the basis of these results, the most recent KDIGO anemia guideline recommends a proactive iron treatment regimen.⁶

PRACTICAL CONSIDERATIONS FOR IRON THERAPY IN PATIENTS WITH CHRONIC KIDNEY DISEASE

Iron therapy in CKD should be guided by iron status tests and clinical considerations and should weigh the potential benefits of avoiding or minimizing blood transfusions, ESA use, and anemia-related symptoms against the risk of potential harm.^{6,10} Iron status testing should be performed monthly in the initial phase of ESA treatment and every 3 months thereafter. The 2012 KDIGO guideline recommended using IV iron if an increase in Hgb concentration or a decrease in ESA requirements is sought and the serum ferritin value is 500 ng/mL or less and TSAT is 30% or less.⁵ The more recent KDIGO guideline recommends initiating IV iron therapy in patients on hemodialysis solely on the basis of laboratory parameters: ferritin ≤500 ng/mL and TSAT ≤30%. For patients with ND-CKD and those undergoing peritoneal dialysis, oral or IV iron should be used if ferritin is ≤100 ng/mL and TSAT ≤40%.⁴⁵ The objective of iron therapy in CKD is to prevent absolute and/or functional iron deficiency because they reduce the effectiveness of ESAs. The response to ESAs can be optimized by the simultaneous use of IV iron, which enables a significant reduction in ESA dosing.⁷⁰⁴ Several studies conducted in the early and late 1990s have demonstrated that IV iron therapy is associated with significant ESA dose reductions.^{636,687,705-713} The DRIVE (Dialysis Patients Response on IV Iron with Elevated Ferritin) study has shown that an intensive IV iron administration protocol (125 mg ferric gluconate with each of eight hemodialysis sessions) can significantly reduce ESA dosing requirements.^{714,715} A Cochrane systematic review provided additional support to the ESA-sparing effects of IV iron.⁶³⁶ Shirazian and associates⁷⁰⁴ noted that the ESA-sparing effects of IV iron were easily demonstrated when iron deficiency prevalence was high and IV iron usage was low. However, they did suggest that most of the benefits of IV iron in reducing ESA use had already been achieved, given that 60% to 80% of U.S. patients on dialysis were being treated with IV iron and that how much additional ESA dose reduction could be obtained with more intensive IV iron regimens was uncertain.

Studies have suggested that ascorbic acid added to the therapeutic regimen of patients treated with ESA and iron had beneficial effects, although neither of these studies was rigorously conducted.^{716,717} Limited evidence has suggested that ascorbic acid may be pro-oxidant and may increase cytokine values.⁷¹⁸ A systematic review and meta-analysis of a limited number of small studies concluded that use of ascorbic acid increases Hgb concentrations, improves

transferrin saturation, and reduces EPO utilization.⁷¹⁹ However, the addition of ascorbic acid to therapeutic regimens is not recommended in KDIGO guidelines.^{5,6}

ANTIINFLAMMATORY AGENTS AND TREATMENT OF ANEMIA OF CHRONIC KIDNEY DISEASE

Medications that directly target hepcidin production in order to treat anemia of CKD are not yet available, but some agents can target inflammatory cytokines responsible for increasing hepcidin production in CKD and have antianemic effects. Antibody-mediated blockade of TNF- α produces an improvement of anemia in rheumatoid arthritis⁷²⁰ and inflammatory bowel disease.⁷²¹ Sotatercept and luspatercept, two novel therapeutic agents that target activin A and possibly other receptors of the TGF- β superfamily, have been under investigation for prevention of vascular calcification and improvement of anemia in CKD.^{722,723} Studies have demonstrated that antiinterleukin-6 (IL-6) therapy improves anemia in patients with CKD. The RESCUE trial analyzed the effects of ziltivekimab, an anti-IL6 antibody, in patients with CKD stages 3 to 5 and systemic inflammation and found that ziltivekimab significantly increased Hgb levels after 12 weeks compared with placebo.⁷²⁴ Additionally, ziltivekimab was associated with significant increases in serum iron levels, total iron-binding capacity, and TSAT.⁵⁴¹ Clazakizumab, a monoclonal antibody that targets the IL-6 ligand, leads to a marked reduction in CRP concentrations in patients receiving hemodialysis with a history of cardiovascular disease and/or diabetes⁷²⁴ and may also improve iron utilization and ESA use.⁷²⁵

EFFICACY AND SAFETY OF ANEMIA MANAGEMENT WITH ERYTHROPOIESIS-STIMULATING AGENTS AND IRON

Improvement in the condition of patients on maintenance dialysis following the advent of rhEPO has been impressive and obviously advantageous. Transfusion requirements declined, iron overload due to previous RBC transfusions gradually resolved, and patients could easily be maintained at Hgb values above those that had to be accepted when RBC transfusions were the only viable option of anemia management. Androgen therapy, which had been associated with significant side effects, could also be eliminated. Because of these obvious benefits, the use of epoetin soon became routine, and the workup for anemia is now considered part of the management program of patients in all stages of CKD (Fig. 53.5). If the workup reveals no reasons for anemia other than EPO deficiency associated with CKD, and in particular has ruled out iron deficiency, ESA therapy provides an option to lessen the degree of anemia in almost all patients. However, despite the apparent advantages, formal evidence of a positive long-term benefit has never been established.

Several lines of indirect evidence have suggested that correcting or ameliorating anemia could reduce or at least mitigate the incidence of left ventricular hypertrophy, a frequent complication in CKD that is clearly associated with poor prognosis (see preceding discussion). Previously, this association of left ventricular hypertrophy, combined with the apparent lack of adverse effects of ESA therapy and the notion that higher Hgb concentrations might improve

HRQOL and physical function, led to an increase in Hgb target values. Furthermore, treatment was expanded to patients not yet on dialysis, in whom anemia is generally less severe than in those on dialysis, based on the concept that avoiding more severe anemia would improve prognosis and quality of life compared with waiting and, then later, correcting more severe anemia.

Unfortunately, however, the true nature of the relationship between long-term reductions in Hgb concentrations and adverse outcomes was not adequately tested in prospective interventional trials for a protracted time course. Few studies have actually compared ESAs against placebo, and those trials, testing two different Hgb target ranges, have usually been inadequately powered before several RCTs were conducted that suggested that normalization of Hgb concentrations with ESAs is associated with limited benefit and relevant harm.

Trials Overview

Since 1989, more than 25 RCTs using ESAs in patients with CKD have been published, in which either different target Hgb concentrations were compared or ESA treatment was compared with placebo. Approximately 50% of these trials were conducted in patients on dialysis and the other half in patients with ND-CKD. Overall, approximately 11,000 patients have been enrolled in these trials, with more than 4000 enrolled in a single study, the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT).⁷²⁶ The number of patients in the other trials varied from fewer than 20 to approximately 1400. Several small trials conducted in 1997 or earlier compared ESA therapy with placebo. Thereafter, only treatment strategies testing two different ESA regimens were performed until TREAT was designed as the first large trial to compare ESA therapy with placebo in patients with diabetes and CKD who were not undergoing dialysis.

Large Randomized Controlled Trials

The U.S. Normal Hematocrit Trial was the first study to test whether normalization of Hgb concentrations improves prognosis of patients on dialysis.⁷²⁷ It was hypothesized that any presumed benefit would be most obvious in patients with cardiac disease; therefore the trial enrolled slightly more than 1200 hemodialysis patients who had congestive heart failure or ischemic heart disease. The target Hct in the higher arm was 42% and, in the lower arm, was 30%. The primary endpoint was a composite of death and first nonfatal myocardial infarction. The study was terminated early after 29 months because more patients in the higher arm had reached the primary endpoint. Although the difference did not reach statistical significance, study termination was recommended because it was obvious that the original hypothesis—that the higher Hct target was of benefit—could not be proven. In addition, the incidence of vascular access thrombosis was significantly higher in the higher target Hct arm. Self-reported physical function scores improved at higher Hct values but, importantly, no significant difference was found between the two treatment arms for this parameter. A later analysis, which included endpoint events that the data safety monitoring committee had not considered when recommending termination of the study, also did not reveal a significant difference,

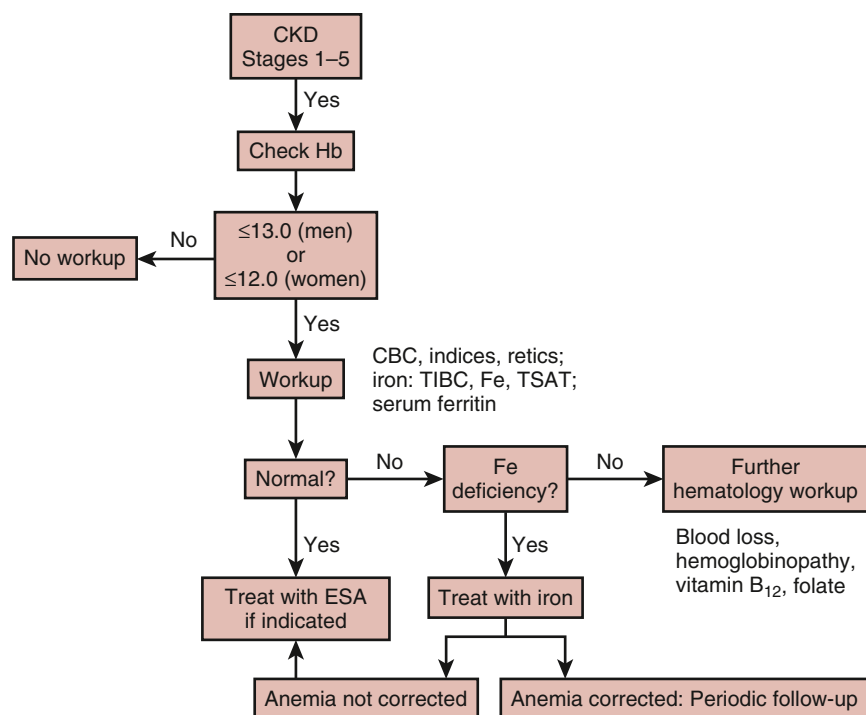


Fig. 53.5 Flow chart for the evaluation of the patient with chronic kidney disease (CKD) and anemia. CBC, Complete blood count; ESA, erythropoiesis-stimulating agent; Fe, iron; Hb, hemoglobin; TIBC, total iron-binding capacity; TSAT, transferrin saturation. (Adapted from Kidney Disease Outcomes Quality Initiative. *Am J Kidney Dis.* 2006;47(Suppl. 3):S1–145.)

and the rates of events during a 1-year follow-up period after study termination were similar in the two treatment arms.⁷²⁸

A second large trial included almost 600 patients new to hemodialysis without symptomatic heart disease and left ventricular dilation.⁷²⁹ Patients were randomly assigned in a double-blind fashion to an Hgb treatment target of either 13.5 to 14.5 g/dL or 9.5 to 11.5 g/dL. The primary endpoint was a change in left ventricular volume index on the assumption that raising the Hgb concentration would prevent the progression of left ventricular hypertrophy. However, changes in left ventricular volume index were similar for the two treatment groups. The only difference among a number of secondary outcomes was a better 36-Item Short Form Health Survey (SF-36) vitality score in the higher Hgb group than in the lower Hgb group. Adverse event rates were similar, except that incidence of skeletal pain, surgery, and dizziness were higher in the lower arm, whereas incidence of headache and cerebrovascular events were slightly higher in the higher arm.

Thus, neither trial provided any evidence in favor of normalization of Hgb concentrations in patients on dialysis. However, because the prognosis, extent of comorbidities, and hemodynamic and metabolic milieu of patients undergoing dialysis are so different from those of patients with ND-CKD, the benefit of anemia correction was further tested in ND-CKD in three larger studies.

The CREATE (Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta) trial was conducted in Europe, Mexico, and Taiwan.⁷³⁰ It enrolled approximately 600 patients with an eGFR of 15 to 35 mL/min/1.73 m² and a Hgb concentration of 11 to 12.5 g/

dL. Patients were randomly assigned to a treatment arm in which epoetin β therapy was started immediately to achieve Hgb concentrations of 13 to 15 g/dL or to an arm in which treatment with epoetin was not initiated before Hgb had dropped to below 10.5 g/dL; the target Hgb in this second arm was 10.5 to 11.5 g/dL. The primary endpoint was a composite of eight cardiovascular events including sudden death, myocardial infarction, acute heart failure, stroke, transient ischemic attack, angina pectoris resulting in hospitalization for 24 hours or more or prolongation of hospitalization, complication of peripheral vascular disease (amputation or necrosis), or cardiac arrhythmia resulting in hospitalization for 24 hours or more.⁷³⁰ The study did not show a significant difference in the time to event between the treatment arms. Some dimensions of HRQOL were improved in the arm with earlier treatment and higher target, but, unexpectedly, time to dialysis was significantly shorter in this treatment arm. One of the limitations of the trial was that the observed event rate was much lower than the anticipated event rate, yielding lower than expected statistical power.

The CHOIR (Correction of Hemoglobin and Outcomes in Renal Insufficiency) study, conducted in the United States, had a similar design but enrolled patients with more comorbidities and yielded different conclusions.⁷³¹ More than 1400 patients with eGFR values of 15 to 50 mL/min/1.73 m² and Hgb concentrations below 11 g/dL were randomly allocated to receive epoetin alfa to achieve one of two different target Hgb values, 13.5 or 11.3 g/dL. The primary endpoint was a composite of death, myocardial infarction, hospitalization for congestive heart failure, and stroke. The trial was terminated when significantly more

patients in the higher Hgb arm experienced at least one cardiovascular event. Compared with the CREATE trial, twofold to threefold higher doses of epoetin were needed in the CHOIR study to achieve and maintain similar Hgb values. Separate analyses of the four components of the combined endpoint revealed trends for more frequent hospitalizations for heart failure and more frequent deaths but no difference in the rates of myocardial infarction or stroke. In addition, there was a trend toward more rapid progression of kidney disease in the higher Hgb target group. Contrary to this finding, a later meta-analysis of 19 trials did not support an effect of higher Hb targets on the progression of CKD, adverse events, or mortality.⁷³² Interestingly, a posthoc analysis of the CHOIR trial has shown that the risks associated with the higher Hgb target were not apparent among subgroups with a higher mortality risk.⁷³³

In contrast to the other four large trials, the TREAT study was designed to test the effect of ESA in comparison with placebo in a sufficiently powered study. More than 4000 patients with CKD (eGFR = 20–60 mL/min/1.73 m²), type 2 diabetes, and a Hgb concentration below 11 g/dL were randomly assigned to receive darbepoetin, with a treatment target of 13 g/dL, or placebo.⁷²⁶ To avoid development of severe anemia in the placebo-treated group, a rescue protocol was established, according to which darbepoetin was administered when Hgb fell below 9 g/dL. The study was double blinded. There were two primary endpoints—a cardiovascular composite endpoint and a renal composite endpoint, including death or initiation of maintenance dialysis. The trial showed no difference in the composite renal or cardiovascular endpoints, but analysis of the components of the primary endpoint revealed a significant, twofold higher risk of stroke in the darbepoetin arm. The numbers of deaths attributed to cancer tended to be higher in the treatment arm, albeit not significantly, and in a subgroup of approximately 350 patients with a history of malignancy, all-cause mortality tended to be higher with significantly more deaths attributed to cancer. These findings were consistent with some findings in ESA RCTs in patients with cancer, which showed higher mortality and more rapid progression of malignancy in patients treated with ESA for chemotherapy-related anemia.⁷³⁴ Patients in the darbepoetin arm of the TREAT study received fewer transfusions and showed a larger mean change in the Functional Assessment of Cancer Therapy: Fatigue (FACT) score.⁷²⁶

Risk-Benefit Relationship and Target Hemoglobin Recommendations

In summary, evidence from well-designed, larger RCTs has indicated that raising Hgb to normal or near-normal values with ESAs does not enhance survival or reduce the rate of cardiovascular events in patients with ND-CKD or kidney failure treated by dialysis but is associated with risk for harm.⁷³⁵ These results are consistent with another large trial in patients with heart failure (RED-HF), many of whom had CKD.⁷³⁶ Almost all studies showed increased rates of thromboembolic events but, for unknown reasons, other risks were not consistent across different studies. Although the CHOIR study, for example, suggested a mortality risk,⁷³¹ this finding was not confirmed in

TREAT.⁷²⁶ Also, a negative impact on the time to dialysis, as found in the CREATE trial,⁷³⁰ was not found in TREAT. TREAT, on the other hand, found an increased incidence of stroke⁷²⁶ and, although another study had previously reported a slightly higher number of strokes in a higher Hgb treatment arm,⁷³⁷ neither the CREATE trial nor the CHOIR study found differences in stroke rates.^{730,731} These inconsistencies may indicate important yet unrecognized factors that determine the side-effect profile of ESAs. Despite intensive investigation, it has not been possible to identify characteristics that distinguish patients in whom stroke developed during ESA therapy in TREAT.⁷³⁸ Whether any of the observed adverse events is related to the actual achieved Hgb values, indirect effects of an increase in erythropoiesis, or direct, Hgb-independent effects of ESAs is unknown.⁷³³

Any benefit of higher Hgb values for HRQOL appears modest, on average, once Hgb concentrations above 10 g/dL are reached. Transfusion rates are lower with higher Hgb values,^{739,740} but it is also clear that attempts to normalize Hgb by no means eliminate transfusion requirements. Moreover, the actual benefit from avoiding RBC transfusions is difficult to determine in individual patients, although the risk of sensitization in prospective transplant recipients should be strongly considered.⁷⁴¹

Whether the balance of the risks and benefits of ESA therapy depends on the patient's responsiveness remains unclear. In treatment protocols driven by a target Hgb range, hyporesponsiveness leads to the use of higher doses and is associated with a greater likelihood of adverse events, but whether ESAs play a causal role remains unclear. A secondary analysis of the CHOIR study has suggested that high ESA doses rather than high Hgb concentrations are associated with poor outcomes.⁷³³ In TREAT, the response to the first two weight-based doses of darbepoetin was a significant predictor of poor prognosis, with patients in the lowest quartile of ESA responsiveness having higher rates of the composite cardiovascular endpoint or death.⁷⁴² However, because "hyporesponders" could be identified only among the treated patients, it is unclear whether their poor prognosis was affected by ESA therapy.

These study data have led to a paradigm shift in anemia management and cautions against overambitious use of ESAs. In patients not undergoing dialysis, it was recommended that a decrease of Hgb values to <9 g/dL be avoided by the initiation of ESA when Hgb is between or even below 9 to 10 g/dL. In general, ESAs should not be used to maintain Hgb concentrations above 11.5 g/dL,⁶ irrespective of whether patients are treated or not treated with dialysis and there is a strong recommendation against intentionally raising Hgb above 13 g/dL.¹⁰ However, some patients may have improvements in quality of life with Hgb values above 11.5 g/dL and may be prepared to accept an increased risk.⁷⁴³

RED BLOOD CELL TRANSFUSION

In the United States, a major change in payment for dialysis and related services has resulted in bundling of payments for laboratory services and IV medications and their oral equivalents. Together with the results from clinical trials and subsequent changes in drug labeling from the FDA and

adapted guidelines, these payment changes have produced measurable reductions in ESA usage and Hgb values and have resulted in higher rates of transfusion in patients on maintenance dialysis.^{35,739,744}

Transfusion rates for U.S. patients on maintenance dialysis were 2.9% in 2011 and 3.0% in 2012. Interestingly, transfusion rates in ND-CKD patients also rose significantly from 2002–2003 to 2008.⁷⁴⁰

Transfusion is associated with the development of alloantibodies and human leukocyte antigen (HLA) sensitization,⁷⁴⁵ which has important negative consequences for donor matching of patient candidates for a renal transplantation. HLA sensitization also increases graft rejection and diminishes graft survival. The current consensus is, therefore, that blood transfusion should be avoided, if possible.⁷⁴¹ In fact, one of the indications for ESA use in CKD is to avoid or at least reduce RBC transfusions in transplant candidates.

Relatively common complications of RBC transfusions are febrile, urticarial, and/or allergic (immediate hypersensitivity) reactions. Less common complications include acute and delayed hemolytic transfusion reactions, hypotensive transfusion reactions, transfusion-associated dyspnea, transfusion-associated circulatory overload (TACO), transfusion-related acute lung injury (TRALI), posttransfusion purpura, and transfusion-associated graft versus host disease. Additional complications of RBC transfusion include potential transmission of known and unknown infectious agents and iron overload.

DISORDERS OF HEMOSTASIS IN CHRONIC KIDNEY DISEASE

BLEEDING AND CHRONIC KIDNEY DISEASE

Excessive bleeding has long been recognized as an important complication of the uremic state.^{746–748} This was particularly true before the advent of dialysis and the availability of rhEPO. Events may be as minor as epistaxis, bleeding with toothbrushing, or easy bruising. More severe, clinically relevant bleeding episodes tend to occur with trauma or after invasive procedures, rather than spontaneously.⁷⁴⁹ Before the availability of routine dialysis, catastrophic gastrointestinal hemorrhage was the major cause of death with uremia.⁷⁴⁶ Bleeding is frequently a predictor of increased mortality risk or complications.^{750,751}

PATHOPHYSIOLOGY

Traumatic disruption of the endothelial lining of blood vessels results in a complex and coordinated response aimed to maintain vascular integrity and prevent bleeding. The first line of defense in hemostasis is by platelets, which specifically interact with ligands exposed as a consequence of endothelial damage. These ligands, which include collagen, fibronectin, laminin, thrombospondin, and von Willebrand factor (vWF), promote adhesion of platelets to subendothelium and their activation. Activated platelets release adhesive ligands stored in their alpha granules, such as vWF, fibrinogen, thrombospondin, fibronectin, and vitronectin and promote the activation of additional platelets by releasing aggregating agents, such as thromboxane A₂ (TXA₂) and adenosine diphosphate (ADP). An occlusive

plug is formed after several minutes by the deposition of platelets on collagen fibers. The surfaces of platelets play an essential role in supporting the coagulation cascade in plasma, which results in the activation of thrombin, conversion of fibrinogen to fibrin, and formation of the fibrin clot. Subsequent cross-linking of fibrin by factor XIIIa further stabilizes the clot. The generation of thrombin further enhances the activation of platelets and upregulates glycoprotein (GP) receptors, such as those for GPIb-IX-V and GPIIb/IIIa. Several systems play an important role in limiting the extent of coagulation activation and thrombus formation. Nitric oxide (NO) and prostaglandin I₂ (PGI₂; prostacyclin) limit the activation of platelets. Tissue factor pathway inhibitor (TFPI), the protein C and protein S system, and antithrombin III (AT III) deactivate coagulation factors at various steps of the coagulation cascade. The fibrinolytic system is also crucial in both limiting the growth of thrombi and promoting their organization and removal. Fibrin digestion is mediated by plasmin, which circulates in plasma as plasminogen, an inactive precursor. Conversion of plasminogen to plasmin is promoted by tissue plasminogen activator (tPA) and inhibited by plasminogen activator inhibitors (PAI-1 and PAI-2).^{752,753}

Several factors contribute to increase the risk of bleeding in patients with CKD (Fig. 53.6).⁷⁵⁴ Enhanced bleeding occurs in uremic patients despite normal or elevated levels of coagulation factors,⁷⁴⁶ suggesting that platelet abnormalities are the primary cause of the bleeding diathesis. In patients with CKD, platelet function is often impaired (thrombasthenia), whereas the number of circulating platelets is generally normal, with a tendency to decline the longer that patients have been undergoing dialysis.⁷⁵⁵ Concentrations of thrombopoietin, the major regulator of megakaryocyte differentiation and platelet production, are elevated in patients who are on maintenance hemodialysis and being treated with rhEPO, but they do not correlate with platelet counts.^{755,756} Evidence for platelet dysfunction includes elevated bleeding time (no longer used clinically),⁷⁴⁶ diminished aggregation responses to ADP and epinephrine,⁷⁵⁷ reduced ristocetin-induced platelet aggregation (measures activities of vWF and its receptor, platelet GPIb),⁷⁵⁸ and prolonged closure time with the Platelet Function Analyzer (PFA-100, Siemens Medical Solutions).^{759–761}

The most consistent abnormality in platelet function in uremia is impaired interaction of platelets with the vascular subendothelium that inhibits platelet aggregation and adhesion.⁷⁴⁶ The cause of this dysfunction is incompletely understood but is likely related to abnormalities of the vessel wall, platelets, and/or plasma constituents. In uremia, endothelial production of NO, a powerful platelet inhibitor, has been noted to be increased,^{762–764} resulting in higher concentrations of cyclic guanosine monophosphate (cGMP) and reduced platelet responsiveness. In uremic rats, treatment with an NO inhibitor partially restores platelet function.⁷⁶⁵ Interestingly, guanidinosuccinic acid (GSA), long postulated to play a role in uremic platelet dysfunction, has been found to upregulate NO production by the endothelium.⁷⁶⁶ Prostacyclin, which is released by the endothelium, is increased in patients with CKD who exhibit increased bleeding times⁷⁶⁷ and probably plays a role in reducing platelet aggregability.⁷⁶⁷

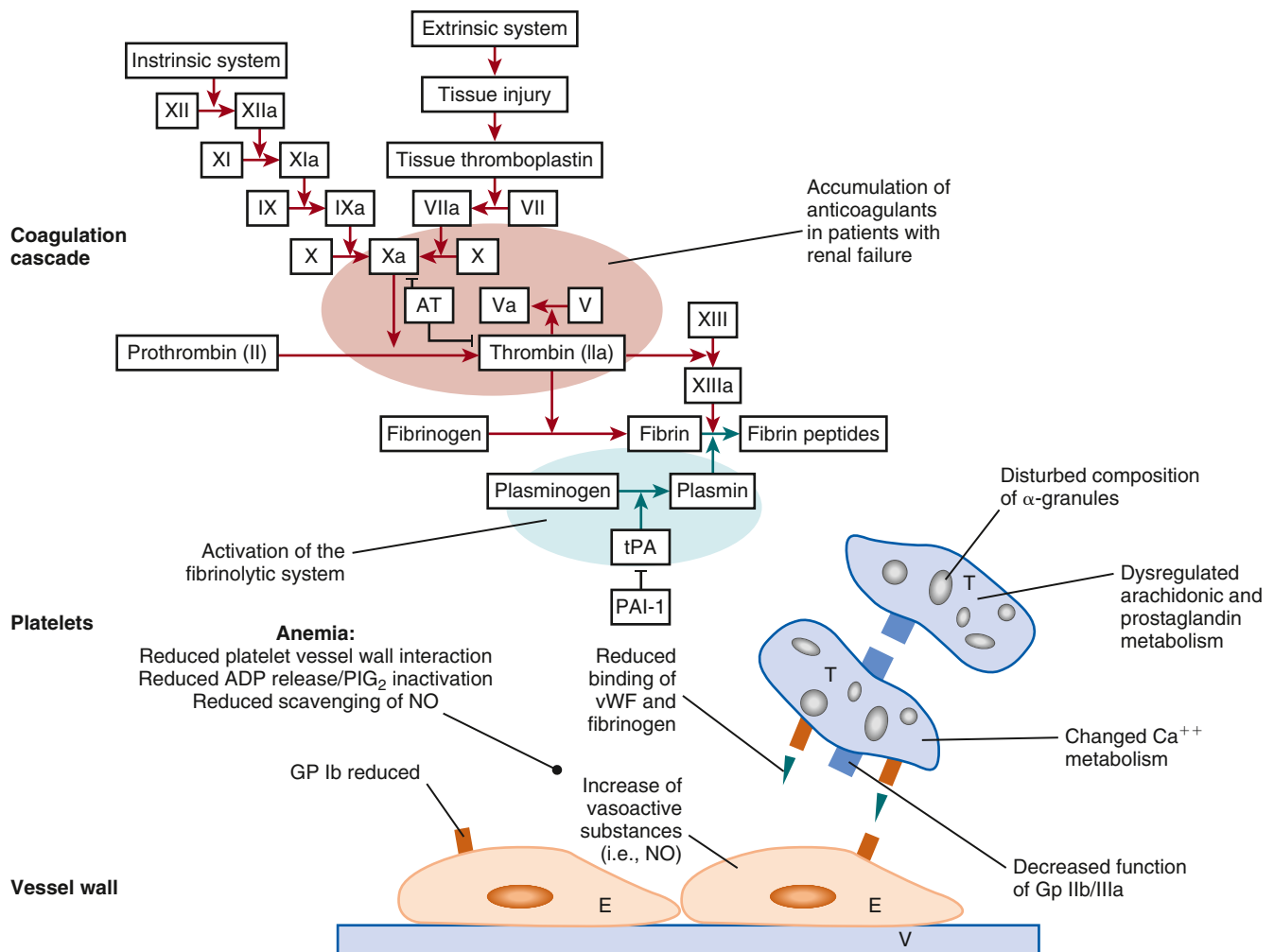


Fig. 53.6 Factors involved in the increased risk of bleeding in patients with renal failure. Roman numerals with or without lowercase letters indicate clotting factors; ADP, Adenosine diphosphate; AT, Antithrombin III; Ca^{2+} , calcium ion; E, endothelium; GP, glycoprotein; NO, nitric oxide; PGI_2 , Prostaglandin I₂ (prostacyclin); T, thrombocyte (platelet); tPA, tissue-type plasminogen activator; V, vessel; vWF, von Willebrand factor. (From Lutz J, Menke J, Sollinger D, et al. Haemostasis in chronic kidney disease. *Nephrol Dial Transpl.* 2014;29:29–40.)

The platelet is intrinsically altered in uremia. For example, the content of serotonin and ADP is reduced in uremic platelet granules.⁷⁵⁷ Secretion of aggregation mediators may also be impaired, although this may be a function of repeated activation during hemodialysis.⁷⁶⁸ Platelet receptors that play a critical role in adhesion to the vessel wall and aggregation, such as those for GPIb and GPIIb-IIIa, do not appear to be significantly reduced in uremia.⁷⁶⁹ However, interaction of the receptors with vessel wall proteins may be abnormal.⁷⁷⁰ In particular, activation of GPIIb-IIIa to facilitate its adhesion to fibrinogen may be impaired.⁷⁷¹ The platelet cytoskeleton may be altered, with diminished actin incorporation and suboptimal intracellular trafficking of molecules.⁷⁷²⁻⁷⁷⁴

Although the platelet itself is not normal in uremia, a more important pathogenic factor in platelet dysfunction may be the effect of uremic plasma on platelet responsiveness. Platelets from normal individuals develop impaired adhesive function on exposure to uremic plasma.⁷⁷⁵ In contrast, platelets from uremic subjects regain some function on exposure to normal plasma.⁷⁷⁵ Certain

molecules with molecular weights that preclude adequate clearance with hemodialysis accumulate in uremia and may contribute to platelet dysfunction.⁷⁷⁶ A variety of toxins, including quinolinic acids and guanidine substances, have been implicated.⁷⁴⁶ Benigni and colleagues⁷⁷⁷ have found that PTH impairs platelet aggregation induced by a variety of substances. Hyperparathyroidism may affect platelet function by elevating intracellular calcium concentrations via channels that are sensitive to calcium channel blockers such as nifedipine.⁷⁷⁸

It is generally accepted that dialysis reduces uremic platelet dysfunction and the risk for bleeding. However, dialysis does not completely eliminate the problem. Moreover, hemodialysis may induce a transient worsening in platelet function. Sloand and Sloand⁷⁷⁹ have measured a variety of indicators of platelet function immediately before and after treatments and noted a transient decrease of platelet membrane expression of GPIb after hemodialysis. Ristocetin responsiveness, which measures GPIb and vWF activities, was impaired after hemodialysis and normalized the day after treatment. Other potential detrimental consequences

of hemodialysis include repeated platelet activation,^{780,781} removal of younger platelets with greater function,⁷⁸²⁻⁷⁸⁴ and impairment of platelet function secondary to an effect of activated leukocytes.⁷⁸⁵

Anemia is an important contributor to uremic platelet dysfunction.⁷⁸⁶ During normal circulation, erythrocytes tend to force the flow of platelets radially, away from the center of flow and toward the endothelial surfaces. When vascular injury occurs, platelets are in closer apposition to the vessel wall, facilitating platelet adherence and activation by vessel wall constituents such as collagen. With anemia, more platelets circulate in the center of the vessel, further from endothelial surfaces, hindering efficient platelet activation.⁷⁸⁶ In addition, anemia may contribute to platelet dysfunction because release of ADP by erythrocytes normally stimulates platelet interaction with collagen.^{787,788} Anemia has also been associated with increased uremic platelet activation markers in patients with CKD stages G1 to G4.⁷⁸⁹ Treatment of anemia may help reverse platelet dysfunction because RBC transfusion^{786,790} and ESA therapy⁷⁹¹ have been found to be beneficial. Anemia is an important risk factor for hemorrhagic stroke—but not ischemic stroke—in patients undergoing hemodialysis.⁷⁹²

The plasma concentrations of the major adhesive proteins, vWF and fibrinogen, are normal in uremia. One study has shown a normal distribution of vWF multimers,⁷⁵⁸ but another reported a reduction in the functional, high-molecular-weight vWF multimers.⁷⁹³ The function of vWF is altered, however, mostly at the level of its interaction with the GPIb/IX-V platelet receptor, a key step in the signaling pathways, which ultimately lead to TXA₂ production.⁷⁹³⁻⁷⁹⁵

Platelet-derived procoagulant microparticles have been described in CKD,^{796,797} but inconsistent and unreliable measurement methodologies have hampered the assessment of their clinical relevance. Circulating endothelial microparticles are increased in CKD, and their baseline values (but not longitudinal changes) are associated with mortality risk.⁷⁹⁸

DIAGNOSTIC STUDIES

Despite abundant evidence that the bleeding time is an unreliable test, with limited value in predicting bleeding complications,⁷⁹⁹ this test has long been used in studies of CKD.⁷⁵⁴ More reliable tests are available, such as standard platelet aggregation and PFA-100, although their value in predicting and managing bleeding complications has been questioned.⁸⁰⁰ Thrombin generation assays may help in assessing both hypocoagulable and hypercoagulable states, but so far there are only limited studies in patients with CKD.⁸⁰¹

TREATMENT

The treatment of patients with CKD experiencing bleeding episodes requires the following:

1. An assessment of the severity of blood loss
2. Hemodynamic stabilization
3. Replacement of blood products as needed
4. Identification of the bleeding source and cause
5. Correction of platelet dysfunction and other factors contributing to the bleeding diathesis

The first four factors are routine components of clinical care and are not discussed further here; the fifth extends from the previous discussion on the pathobiology of uremic bleeding. It should be clear, however, that the intensity of interventions to correct uremic platelet dysfunction hinges on the severity of bleeding.

The first aspect of treatment to correct uremic platelet dysfunction is provision of adequate dialysis. Initiation of dialysis will lead to some improvement in thrombasthenia and bleeding risk.^{757,802} The PFA-100 closure time improves in 25% of patients after a dialysis session.⁸⁰³ No studies have fully elucidated the relative effectiveness of hemodialysis versus other dialytic modalities, but platelet activation measured by CD62 expression was increased by hemodiafiltration, whereas platelet degranulation products were increased in hemodialysis.⁸⁰⁴ In any case, anticoagulation must be minimized. The relationship of dose of dialysis with improvement of platelet function has not been well studied.

Treatment of anemia with ESAs may be the most effective treatment of uremic platelet dysfunction (see earlier). Cases and associates⁸⁰⁵ have found that treatment with epoetin alfa, 40 U/kg IV, results in improvement in several parameters of platelet function as Hgb increases. Others have found the same salutary effect of ESA treatment.^{92,791} Improved platelet function following ESA treatment is most likely related to the associated changes in blood flow, with platelets moving closer to the vessel walls. However, it is also possible that ESA itself may directly affect platelet function. Tassies and colleagues⁸⁰⁶ found that platelet function improves in some patients after epoetin treatment is initiated, before Hgb values increase. They attributed this effect to an increase in young circulating forms of platelets, with improved functional characteristics. Other potential direct beneficial effects of ESA include improved platelet intracellular calcium mobilization,⁸⁰⁷ increased expression of GPIb,^{746,808} and repaired platelet signal transduction.⁸⁰⁹

Desmopressin (1-deamino-8-D-arginine vasopressin; DDAVP) is a synthetic form of antidiuretic hormone that is often used to treat uremic bleeding. The drug has little vasopressor activity but may induce hyponatremia. The mechanism of improved platelet function is not completely known, but enhanced release of larger vWF multimers by endothelial cells probably plays an important role.^{810,811} Other factors may include improved platelet aggregation on contact with collagen⁸¹² and increased concentrations of platelet GPIb/IX, the vWF receptor.⁸¹³ Given the variability of the bleeding time, IV infusion of DDAVP (0.3 µg/kg [0.3 µg/kg subcutaneously]) unsurprisingly produced inconsistent results.^{810,814-816} DDAVP infusion improves platelet function in vitro and increases plasma concentrations for both vWF and factor VIII.⁷⁵⁹ DDAVP may also be administered by the intranasal route, at a dose approximately 10-fold greater than that given intravenously.⁸¹⁷⁻⁸¹⁹ Repeated administrations of DDAVP may result in a diminished response, due to the depletion of the endothelial stores of vWF multimers.^{820,821}

Cryoprecipitate, a plasma product rich in vWF; clotting factors VIII and XIII; and fibrinogen^{822,823} have been suggested to treat uremic bleeding, but supporting evidence is limited.⁸²⁴ Their use should be reserved for

life-threatening bleeding because of the risk for infectious complications and limited availability.

Estrogens improve platelet function in men and women with uremic bleeding.⁸²⁵⁻⁸²⁷ After IV infusion, Livio and associates⁸²⁸ have found the beneficial effect of conjugated estrogens to begin early and last for up to 2 weeks. The mechanism of action of estrogen treatment is not fully known, but it may be related to the inhibition of vascular NO production by decreasing production of its precursor, L-arginine.⁸²⁹ Transdermal estrogens at low doses (≤ 50 $\mu\text{g}/\text{day}$) have been recommended for long-term therapy to balance the hemostatic benefit with thrombotic risks and are preferable to oral ones.^{830,831}

Short-term (6 days) and long-term (3 months) treatments with the fibrinolytic inhibitor tranexamic acid have been associated with a reduction in bleeding time and improved platelet function.^{832,833} Tranexamic acid may also be beneficial in the treatment of acute upper gastrointestinal bleeding episodes.⁸³⁴ Thalidomide is effective for treating for intestinal bleeding from angiodysplasia⁸³⁵ including patients with CKD.^{836,837}

HYPERCOAGULABILITY AND CHRONIC KIDNEY DISEASE

Although bleeding is the most clinically relevant manifestation of the effects of advanced CKD on hemostasis, several lines of evidence indicate the presence of a prothrombotic, hypercoagulable state, which may play a role in the atherosclerotic/cardiovascular complications. Deep venous thrombosis (DVT) more commonly affects CKD patients in younger age, of African-American or Hispanic background, in association with cardiovascular disease and prior surgical interventions.⁸³⁸ The incidence of symptomatic venous thromboembolism is moderately increased in mild-to-moderate CKD (based on eGFR and albuminuria) as shown by a study pooling three European and two U.S.-based community cohorts,^{839,840} as well as by a large population-based study in Denmark⁸⁴¹ and by smaller patient series.⁸⁴² The incidence of pulmonary embolism in people with CKD is not precisely known (it may be particularly common after vascular access procedures—see Chapter 65), but mortality rates for pulmonary embolism are substantially higher in patients on dialysis than in the general population.⁸⁴³ As described earlier, ESA therapy may further increase thromboembolic complications. Vascular calcification and cutaneous necrosis are the key features of calciphylaxis, a serious cutaneous disease affecting 1% to 4% of patients with kidney failure, which has been linked to a hypercoagulable state.⁸⁴⁴

CAUSES OF HYPERCOAGULABILITY IN CHRONIC KIDNEY DISEASE

As outlined in Fig. 53.7, several pathways are altered—toward hypercoagulability and increased risk of thrombosis in CKD.⁷⁵⁴ Activated hypercoagulable platelets have been reported in patients with impaired kidney function,^{845,846} whereas other studies have shown increases in soluble markers of activated coagulation and fibrinolysis.^{847,848} The values of several markers for thrombin activation (prothrombin fragment F1.2 and thrombin-antithrombin complex) and fibrinolysis (D-dimer and plasmin-antiplasmin complex)

are abnormally elevated in dialyzed CKD patients, with erythrocyte membrane phosphatidylserine externalization possibly playing a role in this procoagulant state.⁸⁴⁹⁻⁸⁵² Phosphatidylserine externalization may be mediated by uremic toxins because it improves after dialysis treatment.⁸⁵³

Despite the functional platelet defects described previously, abnormalities in the coagulation cascade and in some of the natural anticoagulant systems (e.g., fibrinolysis) generate a hypercoagulable state, which may facilitate cardiovascular and thrombotic complications in patients undergoing dialysis.^{854,855} Complement activation may take place during dialysis, with increased expression of tissue factor on peripheral blood neutrophils and increased production of granulocyte colony-stimulating factor (G-CSF) resulting in a hypercoagulable state.⁸⁵⁶ Increased prothrombotic markers and platelet activation have been reported following dialysis,⁸⁵⁷ with some variability based on the type of membranes used.⁸⁵⁸ Thromboelastography has shown delayed formation but stronger clots with decreased clot breakdown in patients with CKD, with increased plasma fibrinogen values playing a potential role.⁸⁵⁹ Indolic uremic solutes have also been implicated in the procoagulant phenotype of uremia in animal models⁸⁶⁰ and in patients with CKD.⁸⁶¹ Oxidized plasma albumin has also been implicated in generating a prothrombotic state via CD36-mediated platelet activation.⁸⁶²

PHARMACOLOGIC INTERVENTIONS

Treatment of hypercoagulability may expose patients to additional bleeding complications. A systematic review of bleeding rates in patients with CKD treated with antiplatelet drugs has shown that these agents are effective in reducing thromboses of arteriovenous fistulae and central venous catheters, but not arteriovenous grafts.⁸⁶³ No firm conclusions could be reached about possible increases in bleeding rates in patients treated with a single agent, whereas bleeding risk increased with combination therapy.⁸⁶³ In patients treated for ischemic stroke, the presence of CKD was associated with a twofold increased frequency of clopidogrel resistance (by the VerifyNow P2Y12 Assay, Instrumentation Laboratory, Bedford, Mass).⁸⁶⁴

Atrial fibrillation is a relatively common occurrence in patients with CKD. The optimal approach to prevention of stroke and other embolic complications is unknown. There are concerns that chronic treatment with vitamin K antagonists may worsen vascular calcification.⁸⁶⁵ Warfarin dosing is complicated in CKD by drug-drug interactions, variability in dietary intake, and frequent administration of antibiotics, resulting in poor anticoagulation control.⁸⁶⁶ The risk-benefit balance of warfarin for stroke prevention in advanced CKD is unknown. The increased bleeding risk should be carefully considered,^{867,868} especially in view of conflicting effects observed in older patients with atrial fibrillation and an eGFR <45 to 50 $\text{mL}/\text{min}/1.73^2$, which showed lower mortality despite a higher incidence of hemorrhage and higher or unchanged risk of ischemic stroke.^{869,870} The PROMETHEUS study showed no clinically significant difference in outcomes between prasugrel and clopidogrel treatment in patients with CKD undergoing percutaneous coronary intervention; it confirmed a higher adjusted risk for major adverse cardiac events and bleeding complications in the presence of CKD.⁸⁷¹

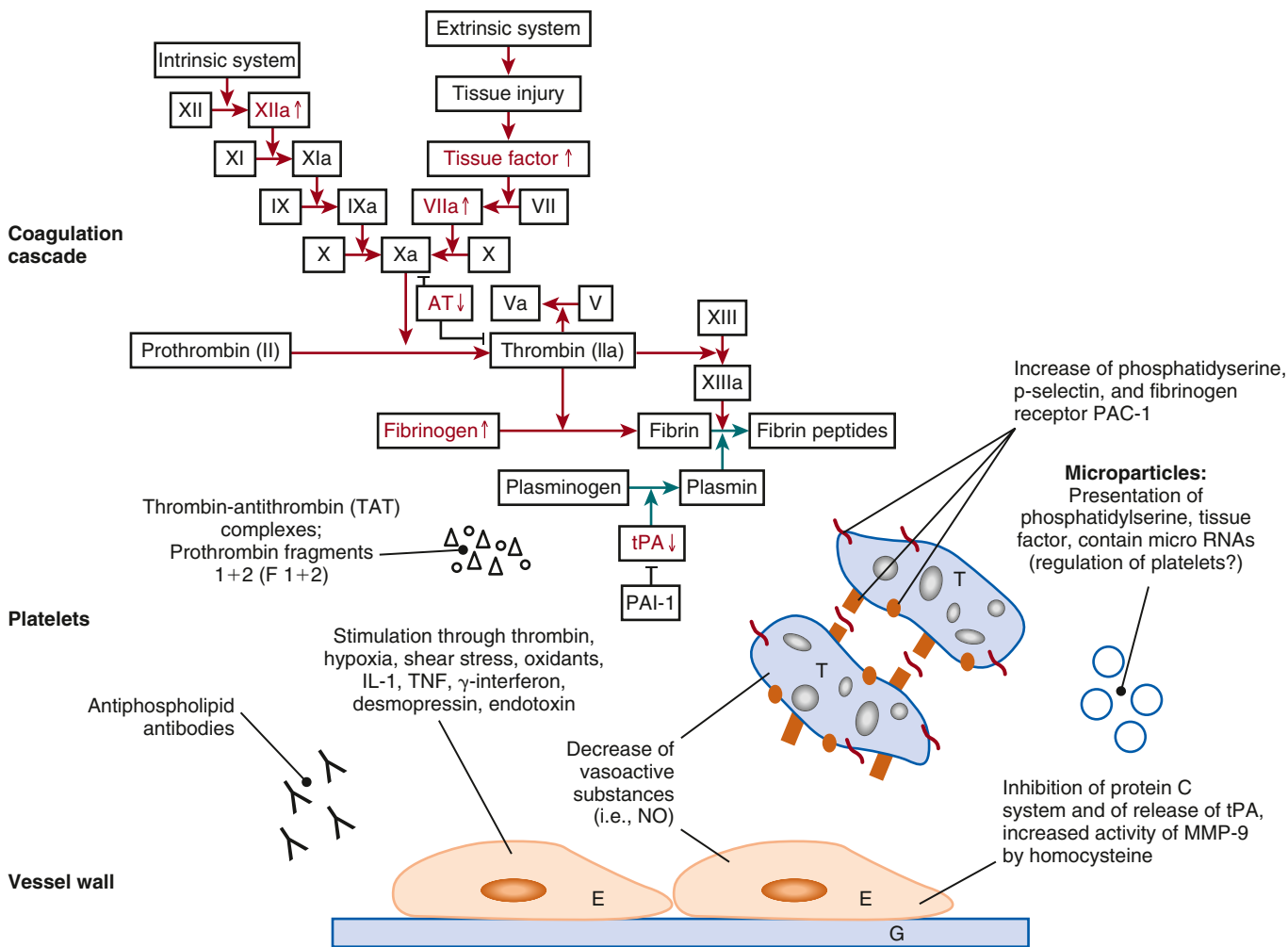


Fig. 53.7 Factors involved in the increased risk of thrombosis in patients with renal failure. Roman numerals with/without lowercase letters indicate clotting factors; AT, Antithrombin III; E, endothelium; G, subendothelial connective tissue; IL-1, interleukin-1; MMP-9, matrix metalloproteinase 9; NO, nitric oxide; PAC-1, monoclonal antibody specific for the activated form of GPIIb-IIIa; PAI-1, plasminogen activator inhibitor-1; T, thrombocytes (platelets); TNF, tumor necrosis factor; tPA, tissue-type plasminogen activator; ↓, decreased; ↑, increased. (From Lutz J, Menke J, Sollinger D, et al. Haemostasis in chronic kidney disease. *Nephrol Dial Transpl.* 2014;29:29–40.)

One study demonstrated no reduction in the risk of stroke but a higher bleeding risk in dialyzed CKD patients with atrial fibrillation who were treated with warfarin.⁸⁷² However, warfarin treatment of atrial fibrillation in patients with CKD who were post myocardial infarction (MI) was associated with lower mortality and a lower incidence of MI and ischemic stroke, without any substantially higher risk of bleeding complications.⁸⁷³ A Danish registry study showed that in high-risk CKD patients, treatment of atrial fibrillation with warfarin results in measurable reductions in all-cause mortality and in hospitalization for stroke and bleeding.⁸⁷⁴ This study was contradicted by a small retrospective study that showed a substantially higher risk of death, bleeding, and MI for patients with CKD stages 3 to 5 including those treated by dialysis when they were also treated with warfarin.⁸⁷⁵ A meta-analysis has shown that warfarin therapy for atrial fibrillation in patients with CKD fails to reduce the risk of stroke and mortality, but it moderately increased the risk of bleeding.⁸⁷⁶ A similar study in patients on maintenance dialysis with atrial fibrillation

and treated with warfarin has shown neither clear benefit nor clear harm.⁸⁷⁷

Direct oral anticoagulants (DOACs) have been approved for use in the general population; studies included a variable fraction (7%–21%) of subjects with impaired kidney function (eGFR <50 mL/min/1.73 m²).⁸⁷⁸ A systematic review and meta-analysis has shown no significant differences in thromboembolic or hemorrhagic complications in CKD patients treated with warfarin or DOACs.⁸⁷⁹ Compared with warfarin, a similar study has shown a significant reduction in bleeding complications in patients with impaired renal function but only for agents with lower renal excretion (<50%; e.g., apixaban, rivaroxaban, edoxaban).⁸⁷⁸ In a primary care setting, bleeding risks and occurrence of stroke were similar in CKD patients with atrial fibrillation treated with an DOAC or warfarin.⁸⁸⁰ Large randomized trials have included patients with reduced kidney function and have shown efficacy for stroke prevention, with no increased bleeding risk.^{867,881} However, due to the lack of specific trials in patients with ND-CKD or those with kidney failure treated

by dialysis, recommendations have been made to use DOAC in CKD stage G3,⁸⁸² not to use DOAC in advanced CKD,⁸⁸³ and to perform proper studies in CKD.⁸⁸⁴

Dabigatran,⁸⁸⁵ a direct thrombin inhibitor, and apixaban⁸⁸⁶ and rivaroxaban,⁸⁸⁷ two factor Xa inhibitors, have been used in patients with CKD. Although low-molecular-weight (LMW) and unfractionated heparin (UFH) can be reversed with the administration of protamine sulfate, the effective reversal of DOACS in the majority of cases is by administration of four factor prothrombin concentrates or the use of specific reversal agents, andexanet alfa for oral factor Xa inhibitors and idarucizumab for dabigatran.⁸⁸⁸⁻⁸⁹⁰ Potpara, Ferro, and Lip have provided recommendations for selection of an oral anticoagulant in patients with atrial fibrillation and CKD.⁸⁶⁷ An essential part of this approach is the proper identification of stroke risk and bleeding risk using established tools such as the CHA2DS2-VASc and HAS-BLED scores, respectively,⁸⁶⁷ although their utility has been questioned by a large Canadian study.⁸⁹¹ The safety of omitting heparinization when dialyzing patients on chronic anticoagulation therapy with vitamin K antagonists has been demonstrated.⁸⁹²

HEPARIN-INDUCED THROMBOCYTOPENIA

Heparin-induced thrombocytopenia (HIT) can be seen in patients undergoing hemodialysis due to their repeated and frequent exposure to heparin.^{893,894} The presence of antibodies to the platelet factor 4–heparin (PF4-H) complex has been associated with arterial and venous thromboses and increased mortality,^{895,896} but other studies have found no correlation between the presence of these antibodies and reduction in platelet counts,⁸⁹⁷ clinical complications,^{894,898} or vascular access thrombosis.⁸⁹⁹ An acute thrombotic event in a thrombocytopenic patient on maintenance hemodialysis or unexpected occlusions of the extracorporeal circuit⁹⁰⁰ should prompt a search for possible HIT. A frequently used clinical guide to HIT probability is the 4Ts scale that includes four criteria: 1. degree of thrombocytopenia, 2. timing of thrombocytopenia onset, 3. presence of thrombosis, and 4. other potential causes of thrombocytopenia. A low score on this scale makes a HIT diagnosis unlikely.⁹⁰¹ For those patients with intermediate or high scores, further screening for the presence of antibodies to the PF4-H complex is performed, usually by enzyme immunoassay.⁹⁰² For those, who have a positive screening test, the functional test for HIT is the gold-standard radiolabeled serotonin-release assay or a similar test. However, the isolated presence of anti-PF4-H antibodies should not by itself lead to a diagnosis of HIT or institution of specific anti-HIT therapies. The presence of oversulfated chondroitin sulfate as a purposeful contaminant of heparins produced in China, which resulted in a large number of adverse events, has also been associated with an increased prevalence of anti-PF4-H antibodies but no thrombocytopenia.⁹⁰³ If the presence of HIT is confirmed on the basis of established criteria,⁸⁹⁴ all heparin-based therapies should be discontinued and the use of direct thrombin inhibitors or factor Xa inhibitors should be considered. In severe cases in patients with CKD, argatroban or danaparoid are used with close monitoring.⁹⁰² For stable patients without thromboses, DOACs can be used until thrombocytopenia resolves. Warfarin should not be considered until the resolution of

thrombocytopenia and neither should prophylactic platelet transfusions. Anticoagulation is extended to 3 to 6 months when thrombosis complicates HIT.⁹⁰²

WHITE CELL FUNCTION IN CHRONIC KIDNEY DISEASE

CKD is accompanied by a chronic inflammatory state of complex pathogenesis, which is believed to be at least, in part, due to an increased generation of oxygen radicals and associated activation of monocytes. Uremic toxins have been thought to predispose to this dysfunctional state, but with no specifically proven connections.^{278,279,904,905} The use of specific dialyzers and dialysates has been associated with intradialytic leukocyte activation and enhanced oxidant stress, which may exacerbate the underlying activated inflammatory state. Activation of platelets adhering to dialysis membranes may contribute to leukocyte activation and the production of ROS.⁹⁰⁶⁻⁹¹² Different types of synthetic dialysis membranes have been shown to induce different degrees of oxidative stress (measured in serum with the surrogate marker malondialdehyde or as ROS).^{913,914}

LEUKOCYTE (MONOCYTE) ACTIVATION

Several studies have shown elevations in markers of leukocyte and monocyte activation in patients undergoing dialysis,^{911,915-917} as well as increased heterotypic aggregation for leukocytes and lymphocytes.⁹¹⁸ Advanced oxidation protein products (AOPPs) carried mostly by serum albumin have been identified in the serum of patients undergoing dialysis.⁹¹⁹ These AOPPs are believed to be end products of protein oxidation. Their concentration is correlated with the severity of uremia, extent of monocyte activation (assessed by serum neopterin),^{92,921} and generation of myeloperoxidase by neutrophils in dialysis patients but not in ND-CKD patients.⁹²² AOPPs can trigger neutrophil activation and respiratory burst, which can be reduced in vitro by N-acetylcysteine.⁹²³ Leukocyte 8-hydroxy-2'-deoxyguanosine (8-OHdG) is a marker of oxidant-induced DNA damage, which is particularly elevated in patients carrying a GST M1 polymorphic dysfunctional variant.⁶⁸⁹ Unique functional features of polymorphonuclear leukocytes collected from patients on hemodialysis include increased polymorphonuclear leukocyte-mediated, transendothelial monocyte migration in vitro and induction of multiple activation markers on cultured human endothelial cells, which depend on superoxide generation.⁹²⁴ The monocytes can contribute to vascular damage as they become macrophages that release inflammatory cytokines in the subendothelial space.

Evidence for leukocyte activation and ROS generation has also been found in patients with ND-CKD.⁹²⁵ Degranulation of neutrophils results in release of a variety of enzymes and proinflammatory mediators^{926,927}; some of these mediators, such as heparanase,⁹²⁸ an endoglycosidase involved in the degradation of extracellular matrix, have been linked to the generation of atherosclerotic lesions; others, such as myeloperoxidase, generate hypochlorous acid, a potent microbicidal and oxidant compound that may play a role in activating monocytes, which produce a whole array of inflammatory cytokines (e.g., IL-6, TNF- α , IL-1 β).

LEUKOCYTE FUNCTIONAL IMPAIRMENT

Granulocytes of patients undergoing hemodialysis exhibit impaired adhesion to fibronectin, especially in conditions of malnutrition.⁹²⁹ Prominent apoptosis is observed in monocytes of patients with CKD,⁹³⁰ and changes in monocyte subpopulations (CD16+) are associated with increased, soluble, proinflammatory markers, such as chemokine (C-X3-C motif) ligand 1 or CX(3)CL1.⁹³¹ Increased production of ROS and the accumulation of toxic products associated with uremia^{928,929} are likely but not yet proven factors in the generation of a dysfunctional immune response in patients with CKD. Phagocytosis by neutrophils is also impaired in patients on hemodialysis and is associated with increased serum endotoxin values.⁹³² Enhanced susceptibility to bacterial or viral infections and reduced response to hepatitis B vaccine and SarsCoV2 vaccines have also been described.^{933,934} Functional abnormalities in monocytes and T lymphocytes,⁹³⁵⁻⁹³⁷ as well as in natural killer (NK) cells,⁹³⁸ have been reported. It has been suggested that such abnormalities may be representative of a myeloid shift away from erythropoiesis, similar to that observed with aging.⁹³⁹ Dialysis membranes designed to reduce immune dysfunction are being developed, with encouraging but still preliminary results.⁹⁴⁰

MARKERS OF LEUKOCYTE ACTIVATION

Elevations in serum CRP or myeloperoxidase have been associated with a higher mortality risk in hemodialyzed patients.⁹⁴¹ Expression studies with oligonucleotide microarray chips have identified distinct patterns of inflammatory and oxidative stress responses in dialyzed patients,⁹⁴² with some evidence suggesting a possible pathogenic role of an impairment of the mitochondrial respiratory system.⁹⁴³ IV administration of vitamin C in a small cohort of patients has produced changes in markers of oxidant stress.⁹⁴⁴ Vitamin C supplementation in hemodialysis is a controversial issue because of the requirements for IV administration, prolonged therapy, and risk of hyperoxaluria.⁹⁴⁵

A better identification of the pathogenesis and clarification of disease modifier genes may allow to design better focused and more personalized treatment approaches for the inflammatory state associated with CKD.^{946,947}

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QUESTIONS

1. A clinically relevant side effect observed following IV administration of ferric carboxymaltose is:
 - a. Hyperphosphatemia
 - b. Hypophosphatemia
 - c. Hypernatremia
 - d. Hyponatremia

Answer: b

Rationale: Transient hypophosphatemia has been reported in patients treated with ferric carboxymaltose, possibly mediated by an increase in FGF23, decreasing tubular reabsorption of phosphate.

2. Chuvash congenital polycythemia, an autosomal recessive disorder endemic in the mid-Volga River region, is due to a mutation in:
 - a. Hypoxia inducible factor (HIF)- α
 - b. Erythropoietin (EPO)
 - c. Erythropoietin (EPO)-receptor
 - d. Von Hippel-Lindau (VHL) protein

Answer: d

Rationale: Chuvash polycythemia occurs in individuals who are homozygous for a specific mutation in *VHL*. The genes encoding *HIF- α* , *EPO*, and *EPO-R* are not mutated in Chuvash polycythemia.

3. Hepcidin is a peptide hormone produced by the liver, which affects iron metabolism by:
 - a. Negatively regulating both duodenal iron absorption and iron release from macrophages
 - b. Positively regulating both duodenal iron absorption and iron release from macrophages
 - c. Negatively regulating only duodenal iron absorption and not iron release from macrophages
 - d. Positively regulating only duodenal iron absorption and not iron release from macrophages

Answer: a

Rationale: Hepcidin binds and leads to internalization and degradation of the iron exporter ferroportin, which is

located on all cell types. Like all cells in the body, the duodenum and macrophages export iron via ferroportin, and hepcidin negatively affects their iron export function. Hepcidin has no positive effects on iron export.

4. Identify the test which most closely reflects the functional availability of iron for erythropoiesis in CKD.
 - a. Serum ferritin
 - b. Bone marrow iron stain
 - c. Serum transferrin receptor
 - d. Reticulocyte hemoglobin content

Answer: d

Rationale: Low serum ferritin found in iron deficiency can be increased by inflammation; bone marrow iron staining is difficult to quantify, especially when decreased but not absent; serum transferrin receptor is elevated in both iron deficiency and erythropoietic expansion. Reticulocyte hemoglobin content is decreased in iron deficiency and inherited hemoglobinopathies, but the latter are relatively uncommon and usually recognized in childhood.

5. Please identify the correct statement regarding the bleeding diathesis observed in patients with CKD.
 - a. The platelet itself is normal in uremia.
 - b. The interaction of platelets with the vascular endothelium is impaired in uremia.
 - c. Uremic platelet dysfunction is independent of anemia.
 - d. Adhesive proteins like von Willebrand factor (VWF) and fibrinogen are decreased in uremia.

Answer: b

Rationale: Platelet function is impaired in uremia and worsened by anemia, which rheologically moves platelets away from vascular walls where they interact with the endothelium to promote hemostasis. VWF and fibrinogen are normal in uremia.